Morse Micro



Morse Micro OpenWrt Web UI User Guide

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Table of Contents

1 Overview 4
1.1 Glossary of Acronyms 5
2 Important Security Notice 6
2.1 HTTPS-Only Access 6
2.2 SSH Disabled by Default 7
2.3 Software Updates 7
3 Device Setup 8
3.1 EKH01 8
3.2 EKH03 10
3.3 EKH19 13
4 Standard Setup Scenarios 15
4.1 Standalone Access Point with Client Devices 15
4.2 Using HaLow as a Virtual Wire (Layer 2 bridge) 15
4.3 Non-Standalone Access Point with Routing 16
5 Configuration of Operating Modes 17
5.1 Default Web UI IP Addresses 17
5.2 Supported Browsers 17
5.3 Initial Setup 18
5.4 Standalone AP and STA 19
5.5 Virtual Wire - Layer 2 Bridging 23
5.6 Non-Standalone AP with Routing 24
5.7 Setting a Custom Static IP 24
5.8 Using DPP QR Code 26
5.9 Using DPP Push Button 28
5.10 11s Mesh 29
5.11 EasyMesh 34
5.12 Monitor Mode 37
5.13 Native S1G (beta) 38
6 Performance Testing and Evaluation 39
6.1 Wavemon and Ping Testing 39
6.2 Setting Up iPerf3 Traffic Testing 40
6.3 Range Testing 41
6.4 Video Streaming 44
6.5 HaLow Diagnostics 47
7 Page Descriptions 50
7.1 Home 50

7.2 Upgrade 50
7.3 Quick Config 51
7.4 Status 54
7.5 System 55
7.6 Services 56
7.7 Network 57
7.8 Statistics 59
7.9 Help 60
8 Additional Configuration Parameters 61
8.1 Disable AMPDU 61
8.2 Fragmentation Threshold 61
8.3 Station Inactivity Limit 62
8.4 Unified Scaling Factor / Unscaled Interval 62
8.5 DTIM Interval 63
8.6 Beacon Interval 64
8.7 BSS Color 65
8.8 Thin LMAC 66
8.9 Other HaLow settings 68
8.10 morse_cli 68
8.11 Dynamic Channel Selection (DCS) 68
9 UI Configuration Architecture 70
10 Troubleshooting 72
10.1 Restoring Factory Settings 72
10.2 Firmware Update Failures 74
11 Revision History 75

1 Overview

Thank you for choosing to evaluate Morse Micro 802.11ah HaLow for use in your application. This guide will get you started using the kit and evaluating the 802.11ah technology. It is primarily intended for users of the web UI but will mention other configuration methods for reference.

The Morse Micro web UI provides a graphical interface for viewing and modifying device configurations, particularly the operating mode and HaLow radio parameters. The interface is available on EKH01, EKH03 and EKH19 evaluation kits and is based on OpenWrt's standard LuCI interface.

The contents of this document can be summarised as follows:

- The Device Setup section of this document provides a brief description on how to set up the hardware.
- Standard Setup Scenarios outlines the basic device configurations that might be used for evaluation.
- Configuration of Operating Modes explains how to configure a system for the first time using the Morse Micro web UI.
- The Performance Testing and Evaluation section describes ways to test the performance of Wi-Fi HaLow using Wavemon, iPerf3, the Range Test tool and video streaming capabilities respectively.
- Page Descriptions outline some of the available UI screens and tools.
- Additional Configuration Parameters provides tips that are not usually required but may be useful in some situations.
- UI Configuration Architecture provides a functional explanation, including how UI configuration propagates through the system to effect changes.
- Finally the Troubleshooting section provides some advice for common problems.

1.1 Glossary of Acronyms

Acronym	Full Form	Definition
AP	Wi-Fi Access Point	The device that creates and manages the Wi-Fi network which stations / clients connect to, similar to your home router.
STA	Wi-Fi Station / Client	Any Wi-Fi-enabled device (like a phone or laptop) that connects to an AP.
UI	User Interface	The web-based graphical interface that lets you interact with your evaluation kit using buttons, forms, and menus. It's designed to be the more intuitive way to use Morse Micro OpenWrt.
CLI	Command Line Interface	A text-based way to control evaluation kits using commands, intended for advanced users and accessed via SSH or the Terminal page.
UCI	Unified Configuration Interface	The system OpenWrt uses to manage configuration through files in `/etc/config/`.
SSID	Service Set Identifier	The name of the Wi-Fi network.
SSH	Secure Shell	A secure way to log in to the CLI remotely.

Table: Acronyms and Definitions

2 Important Security Notice

There have been some significant updates to the Morse Micro OpenWrt software in order to remain compliant with international cybersecurity requirements.

2.1 HTTPS-Only Access

The web interface from Morse Micro OpenWrt 2.11 onwards, is accessible only via HTTPS and any attempt to connect using HTTP will be automatically redirected. When you first connect, your browser may display warnings about the page being insecure. This is expected, because the device uses an encrypted HTTPS session without a signed certificate. You must proceed through these warnings in order to continue accessing the device. This is safer than HTTP as all traffic will be encrypted.

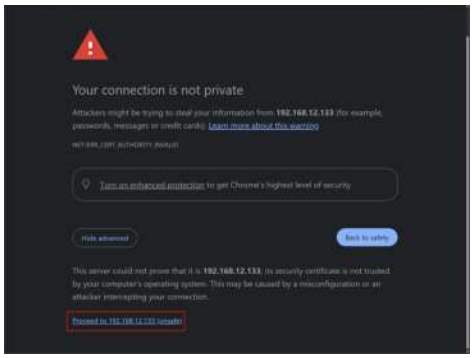


Figure: Chrome HTTPS warning page

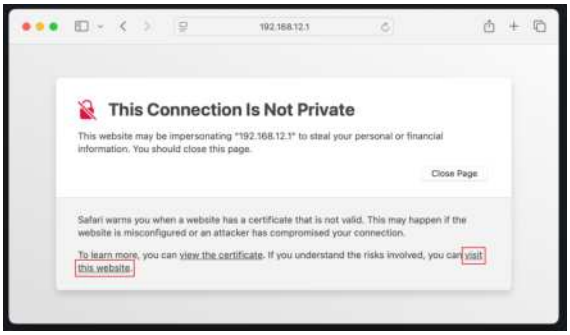


Figure: Safari HTTPS warning page

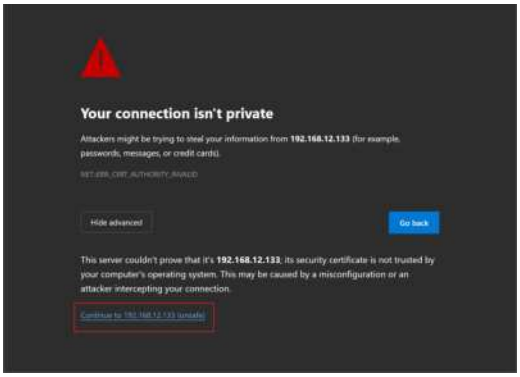


Figure: Edge HTTPS warning page

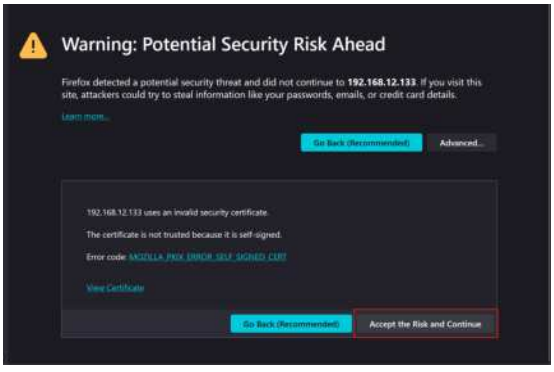


Figure: Firefox HTTPS warning page

2.2 SSH Disabled by Default

To enable SSH, navigate to the **SSH Access** tab on the Administration Page.

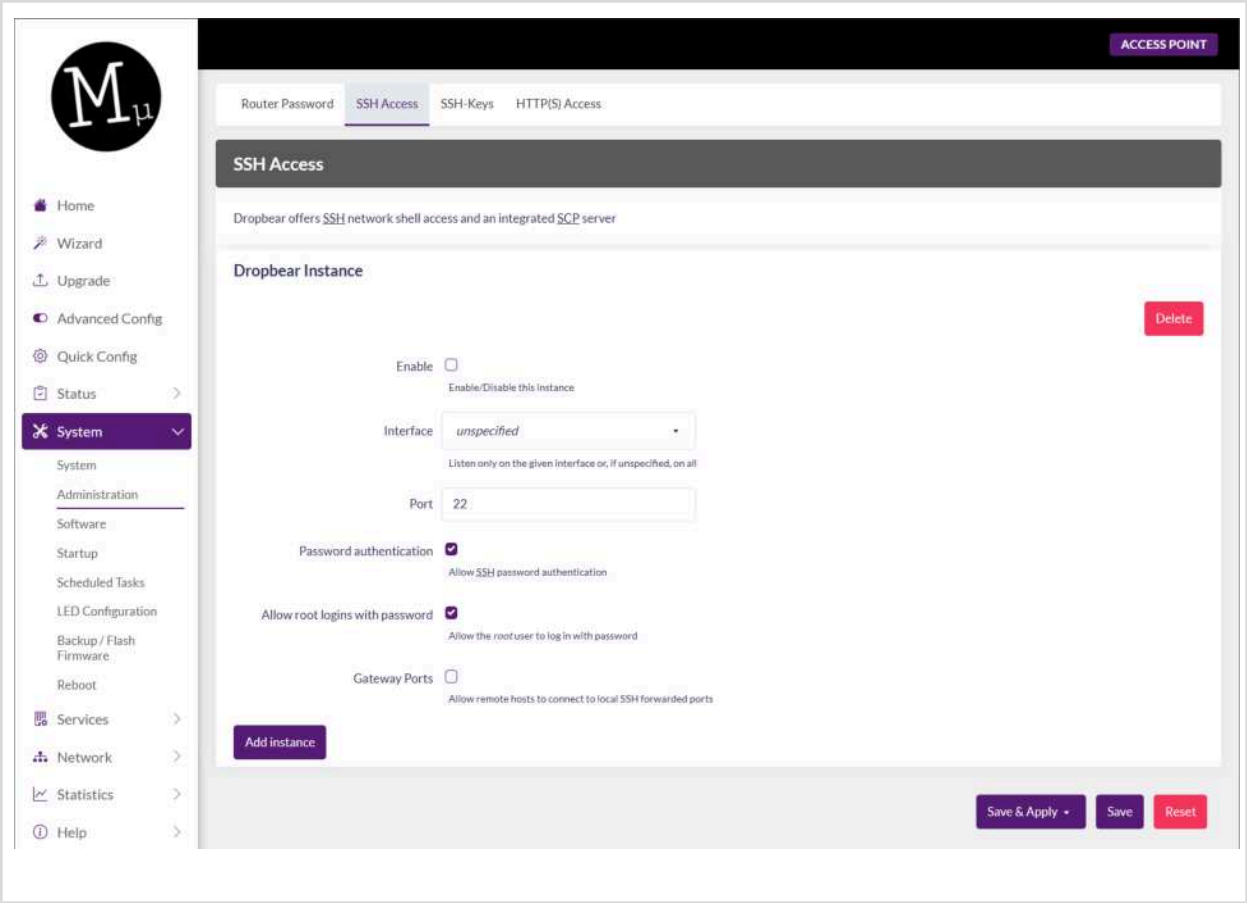


Figure: The SSH Access tab on the Administration page

2.3 Software Updates

To maintain compliance with evolving cybersecurity requirements, updates for this device may be released more frequently and it is strongly recommended to regularly check for new updates via the Upgrade page.

3 Device Setup

Below is a brief description of the hardware and physical setup steps for the relevant evaluation kits. This guide applies to all variants of the evaluation kits described below, including those based on different Morse Micro HaLow chips such as the MM6108 and MM8108.

3.1 EKH01



Figure: EKH01 diagram 1



Figure: EKH01 diagram 2

Physical Feature	Description
microSD card	This contains the device firmware.
Status LEDs	The red LED indicates power, the green LED indicates HaLow network activity.
USB Type-C	Port for supplying power to the EKH01. The kit includes an AC adaptor that converts mains power to 5V for the EKH01 via the USB-C connector.
Micro HDMI	Not typically used.
Headphone Jack	Not typically used.
USB Ports	USB-A ports for connecting peripherals and USB to serial adapter. Any of these ports can be used for serial console access, but note the cable must be plugged in at boot time to be detected. The serial console operates at 115,200 baud 8N1 by default.
Ethernet	Ethernet port for either a LAN connection (e.g. a laptop) or an upstream WAN connection (e.g. gateway router).
(Optional) Camera	The device may include a camera depending on the kit version ordered.

Table: EKH01 physical feature descriptions

3.1.1 Basic Setup

1. Connect the antenna to the HaLow antenna connector on top of the unit.
2. Optional - connect an RJ45 Ethernet cable to the Ethernet port if required.
3. Optional - connect a USB-serial cable to any of the USB ports if required for debugging. This is not usually required.
4. Once power is supplied, it should take the device around 60 seconds to boot up and be operational.

3.2 EKH03

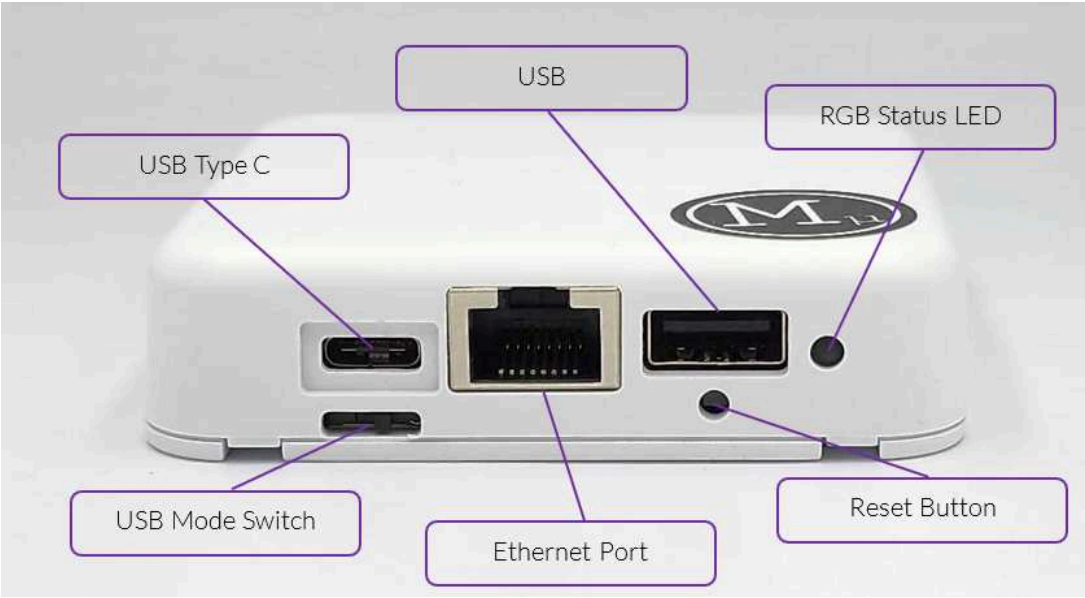


Figure: EKH03 diagram 1



Figure: EKH03 diagram 2

Physical Feature	Description
USB Type C	Powers the board and can function as an Ethernet-to-USB adapter if the USB mode switch is in the leftmost position.
USB	Can be used for connecting peripheral devices to the EKH03 and provides a UART interface which can be accessed via a 3.3V TTL serial USB cable. For advanced users, UART access is also available through header pins on the internal PCB.
RGB Status LED	Is a multi-color LED that is used to indicate the status of the device (see LED Behaviour for details).
USB Mode Switch	Selects whether to use the USB-C or Ethernet port for the LAN connection. The direction of the switch determines which port is selected (left - USB-C for Ethernet, right - Ethernet port for Ethernet).
Ethernet Port	For either a LAN connection (e.g. a laptop) or an upstream WAN connection (e.g. gateway router).
Reset Button	Can be used to reset the device. Pressing and holding the button for 5 seconds (until flashing green) will reboot the device. Holding for more than 10 seconds (until flashing yellow) will trigger a full factory reset of the device, returning it to the default configuration. This button is also used for DPP push button connections. For more details, see Restoring Factory Settings and Using DPP Push Button.
2.4 GHz Wi-Fi	Provides an access point which is brought up on the EKH03 by default. It is also automatically bridged to the Ethernet interface.

Table: EKH03 physical feature descriptions

3.2.1 Basic Setup

1. Connect the provided antenna to the HaLow Antenna Connector shown.
2. Plug the power adapter into the USB-C connector on the EKH03. The included USB-C to USB-A cable can also connect the EKH03 to a power source, such as a laptop or phone charger (via the USB-C connector).
3. Once the device is powered, the RGB LED will display the status of the device (see table).

3.2.2 LED Behavior

Status	LED Color	Graphical Example
Booting stage 1 (bootloader)	Yellow (solid)	
Booting stage 2 (linux)	Green (flash 1.5 per sec)	

Status	LED Color	Graphical Example
Fully booted	Green (solid)	
DPP running	Purple (flash 1 per sec)	
DPP failed (remains for 5 secs)	Purple (flash 5 per sec)	
Fully booted and HaLow connected	Purple (solid, flash activity)	
Error	Red (flash 10 per sec)	
Rebooting	Green (flash 1.5 per sec)	
Upgrading	Blue (flash 1.5 per sec)	
Factory resetting	Yellow (flash 5 per sec)	

Table: EKH03 LED behavior

3.3 EKH19

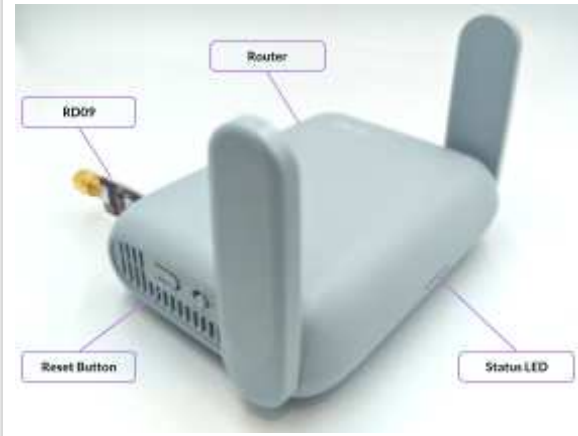


Figure: EKH19 diagram 1

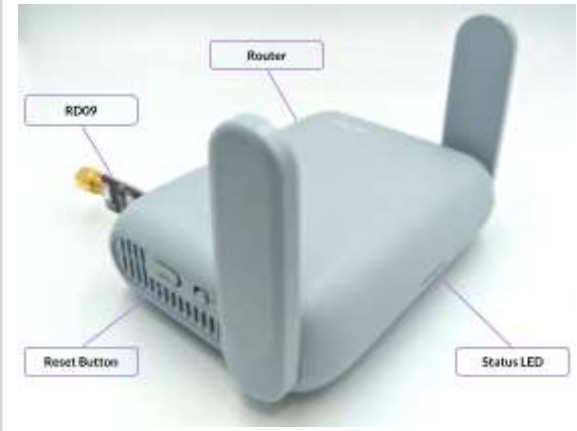


Figure: EKH19 diagram 2



Figure: EKH19 diagram 3

Physical Feature	Description
RD09	Wi-Fi HaLow USB dongle.
Router	A host for the RD09.
Reset Button	Used to reset the device. Pressing and holding the button for longer than 5 seconds (until rapidly flashing blue) will trigger a full factory reset.
Status LED	A dual-color LED that is used to indicate the status of the device (see Basic setup for details).
USB Type C	USB-C port for supplying power to the EKH19. The kit includes an AC adapter that converts mains power to 5V for the EKH19 via the USB-C connector.
Ethernet Ports	Despite the markings, either port can be configured as either a LAN connection (e.g. a laptop) or an upstream WAN connection (e.g. gateway router).
2.4 & 5 GHz Wi-Fi	Provide access points which are brought up on the EKH19 by default. They are all automatically bridged together, including the ethernet interfaces.
Firmware LED	A green LED indicating that the RD09 has loaded correctly.
Association LED	A green LED which will light up only if the HaLow network has been configured as a station and has successfully associated with an Access Point.
Traffic LED	A red LED which will blink to indicate HaLow network traffic.

Table: EKH19 physical feature descriptions

3.3.1 Basic Setup

1. Plug the provided RD09 Wi-Fi HaLow USB dongle into the router's USB Type-A port.
2. Connect the antenna to the HaLow antenna connector on the RD09.
3. Optional - connect an RJ45 Ethernet cable to the LAN Ethernet port if required.
4. Once power is supplied, the status LED will begin flashing blue. Once the router has fully booted it will become solid blue. This process normally takes about 40 seconds.

4 Standard Setup Scenarios

4.1 Standalone Access Point with Client Devices

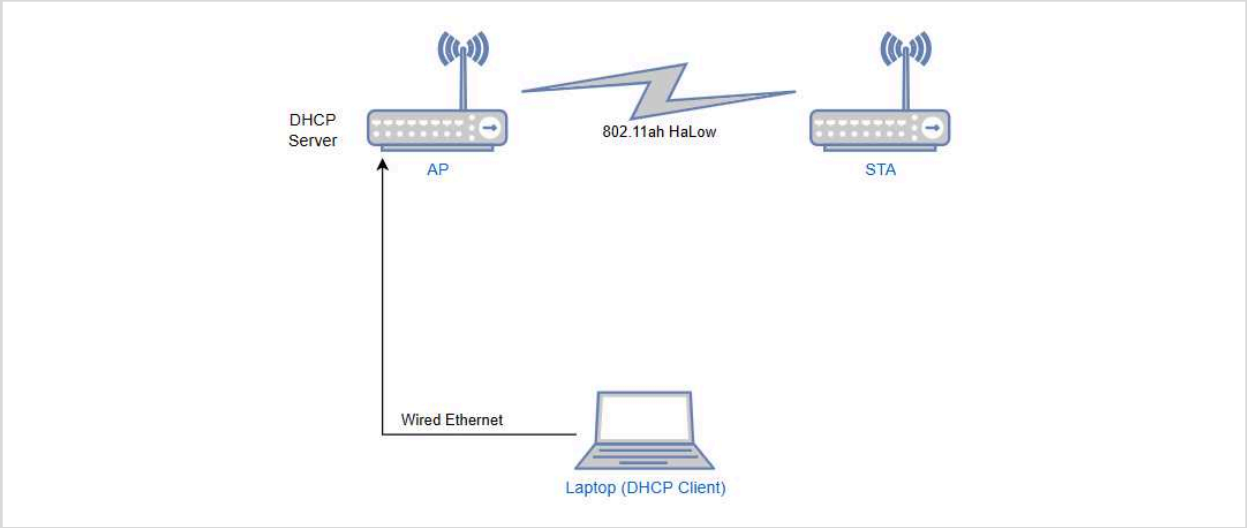


Figure: Standalone Access Point with Client Devices

This configuration is typically used for standalone testing of a HaLow connection. It is useful in closed network scenarios where connected devices do not require access to external networks. The key here is that the traffic will only go between the AP and STA and is completely isolated from any external networks. If you're unsure which setup to use, start with this one.

4.2 Using HaLow as a Virtual Wire (Layer 2 bridge)

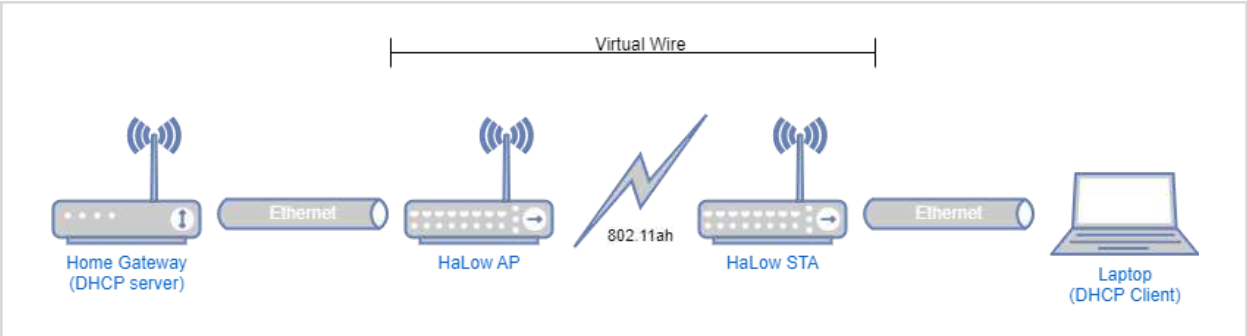


Figure: Using HaLow as a Virtual Wire (Layer 2 bridge)

In this configuration HaLow is transparent to the rest of the devices in the network. The HaLow link is used as a means of providing a 'virtual' Ethernet connection between two points where it may not be practical to run a physical cable. This scenario is useful as a simple way to test HaLow with real-world traffic by introducing it into an existing network without having to adjust the configuration of the non-HaLow devices.

4.3 Non-Standalone Access Point with Routing

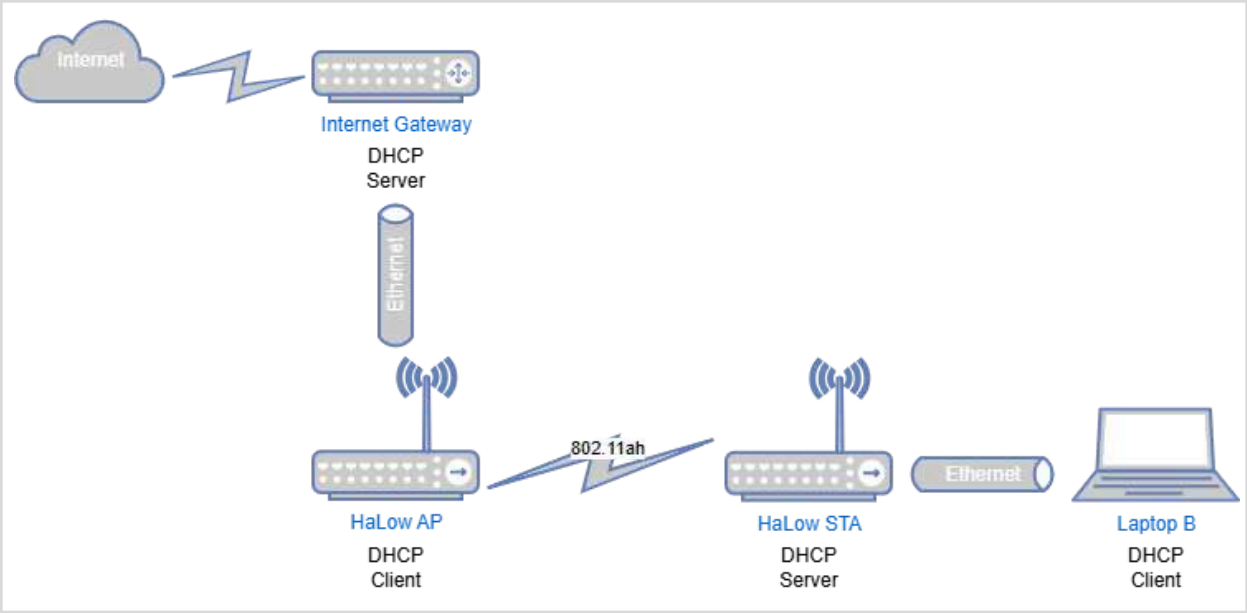


Figure: Non-Standalone Access Point with Routing

This scenario is a more complicated version of the one above, where rather than using bridging to simplify the setup, each device is a router with its own DHCP server and local network. This allows for a more complex network configuration, but is more difficult to set up. It is also robust in that if the HaLow link goes down, the station will still have an IP address and the web UI will be reachable.

This scenario is useful for evaluating the HaLow device's ability to handle traffic flows at Layer 3, but places more load on the CPU. Unless you have a specific reason to use this approach, bridging is a simpler alternative.

5 Configuration of Operating Modes

Evaluation kits are dispatched in a default configuration and this guide assumes the devices will be used starting in that state. If the kits have been used previously you may need to reset them to their default state before following the steps below.

5.1 Default Web UI IP Addresses

The following table lists the default IP addresses used to access the Web UI on the evaluation kits. This Web UI is the primary graphical interface provided by the Morse Micro OpenWrt system, and is referenced throughout this guide for configuration and management.

When connecting to a Morse device over Ethernet, your host device (e.g., laptop or mobile phone) should automatically receive an IP address as a DHCP client. You may need to manually configure your host with a static compatible IPv4 address on the same subnet (e.g., 10.42.0.X or 192.168.8.X). Refer to your device's network settings to set a static IP if the connection fails to load the interface.

Evaluation Kit	Ethernet Interface IP	HaLow Interface IP
EKH01	10.42.0.1	192.168.12.1
EKH03	10.42.0.1	192.168.12.1
EKH19	192.168.8.1	192.168.12.1

Table: Default IP addresses

5.2 Supported Browsers

The web UI has been tested and verified to work with up to date releases of the following browsers:

- Google Chrome
- Firefox
- Microsoft Edge
- Apple Safari

5.3 Initial Setup

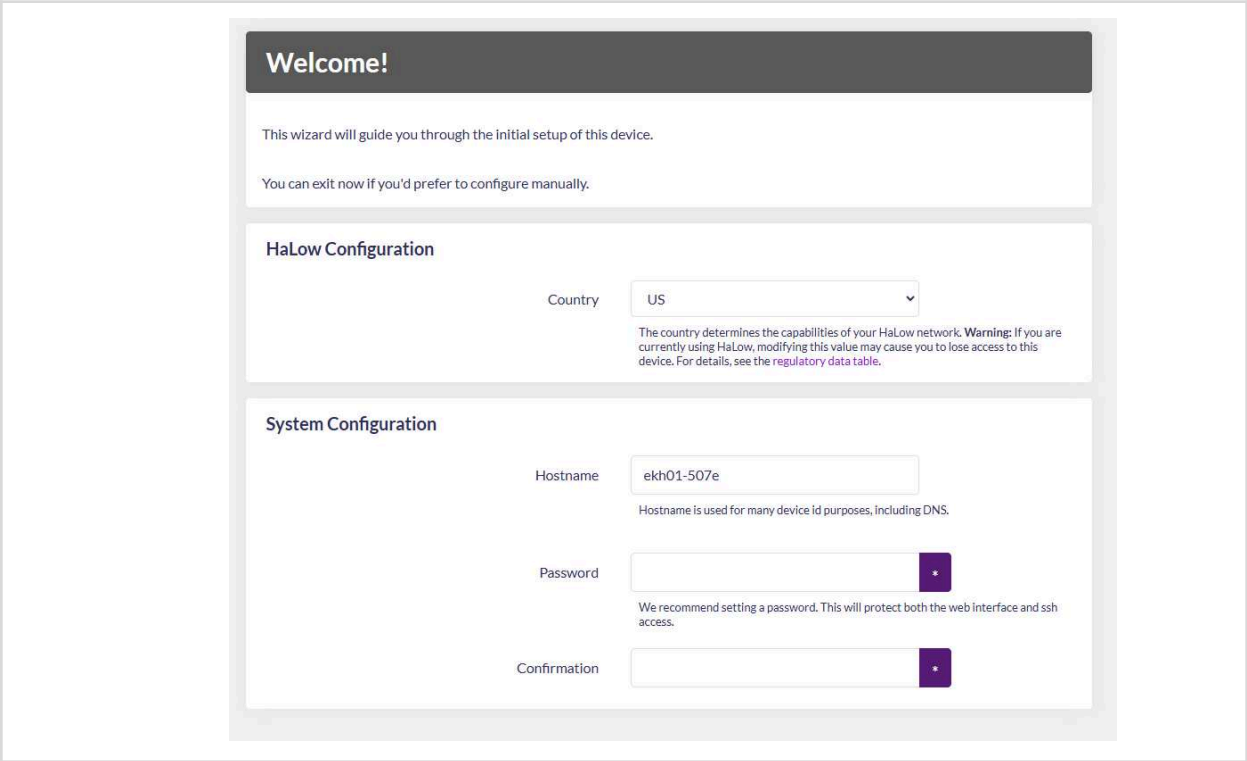


Figure: Initial configuration landing page

1. Connect your laptop to the Morse Micro HaLow device via an Ethernet cable.
2. Ensure that the Ethernet interface on the laptop is configured as a DHCP client (this is usually default, so often no change is required).
3. Open a browser and navigate to the default web UI IP address. You should see a screen like the one pictured.
4. Select the **Country** (for regulatory requirements) and optionally set a unique **Hostname** for the device and a **Password** (which secures SSH and web access):
5. Click **Apply** in the bottom right.

The next screen will present an option to configure the device either as a standalone AP/STA or as part of a mesh network using 802.11s mesh or EasyMesh. For first time users, the Standard Wi-Fi HaLow wizard is the best option to start with. Mesh configurations allow multiple APs to be linked in order to provide an even wider coverage area.

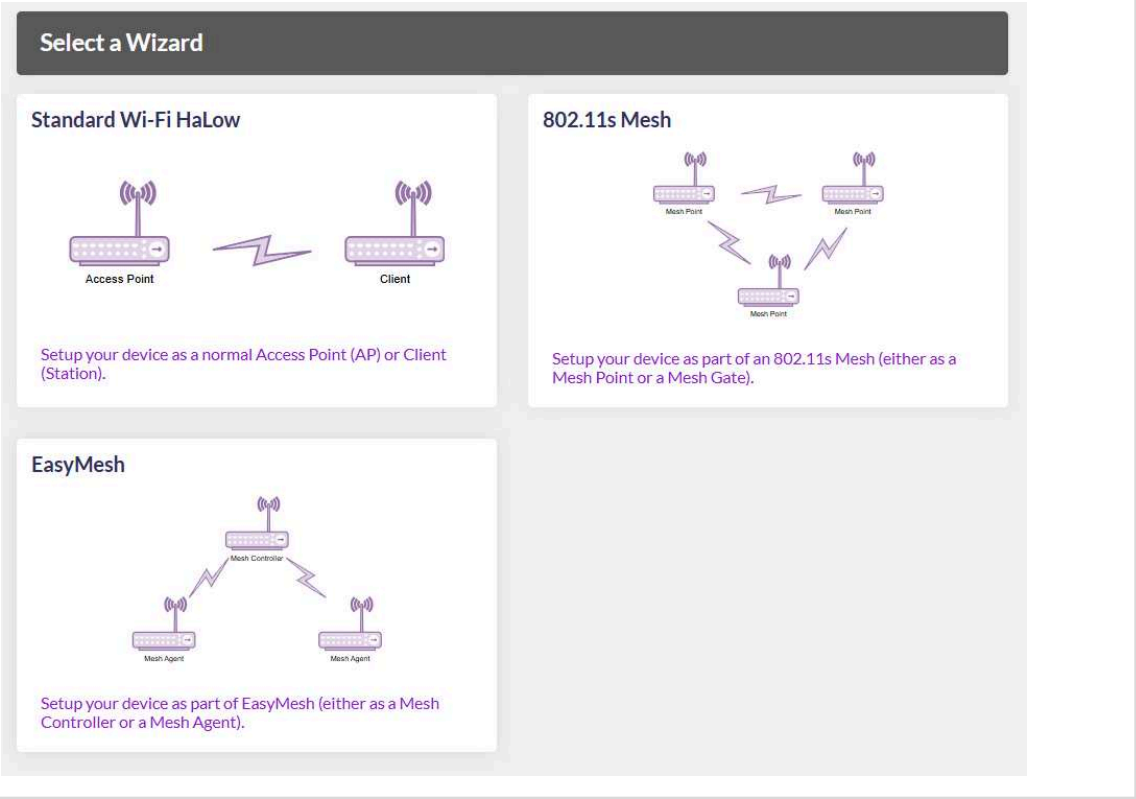


Figure: Wizard selection page

5.4 Standalone AP and STA

This section outlines how to configure the AP and STA per the scenario defined in Standalone Access Point with Client Devices.

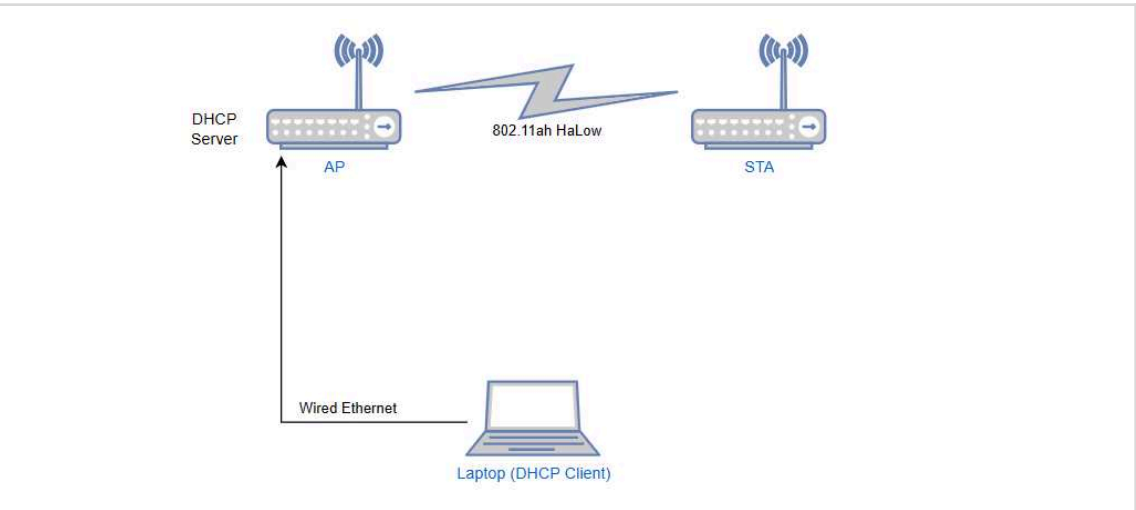


Figure: Standalone AP and STA

5.4.1 Access Point Configuration

1. Follow the steps in Initial Setup to connect to the device and set the region.
2. For Mode selection, choose **Access Point**:

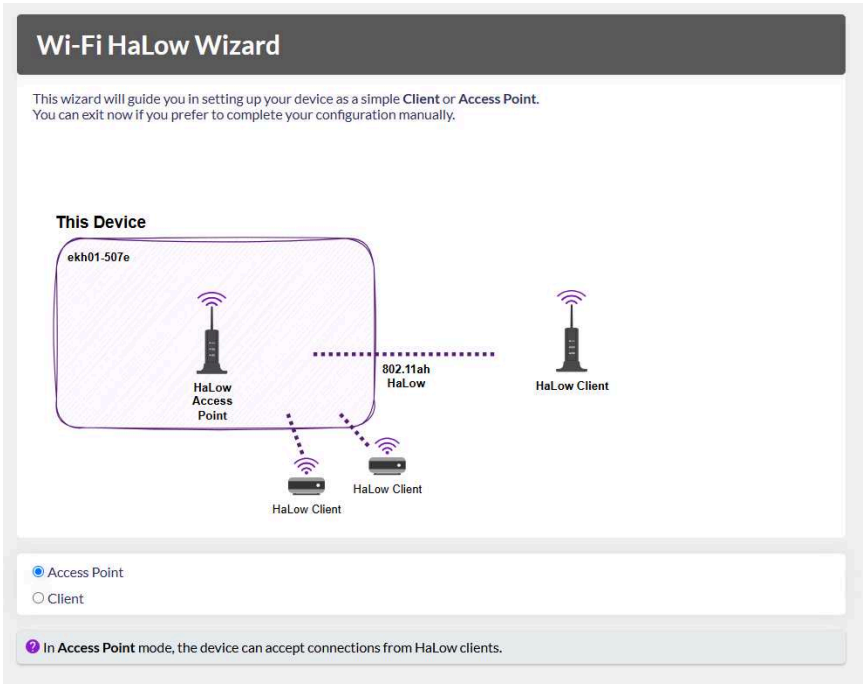


Figure: Standalone AP and STA (Access Point selection)

3. Click the **Next** button at the bottom right to go to the next page.
4. On the following pages:
    a. **Setup HaLow Network - AP** has a default SSID and passphrase which you can change if you wish. The Width and Channel options can be left as default. Then click **Next**.
    b. **Upstream Network** should be set to **None**. Click **Next**.
5. You can then **Apply** your configuration on the final page.

5.4.2 Station Configuration

1. Disconnect your laptop from the Ethernet interface of your AP (from Access Point Configuration above) to prevent IP address conflicts with the client.
2. Follow the steps in Initial Setup to connect to the device and set the region.
3. For the Mode selection, choose **Client** and click **Next**.

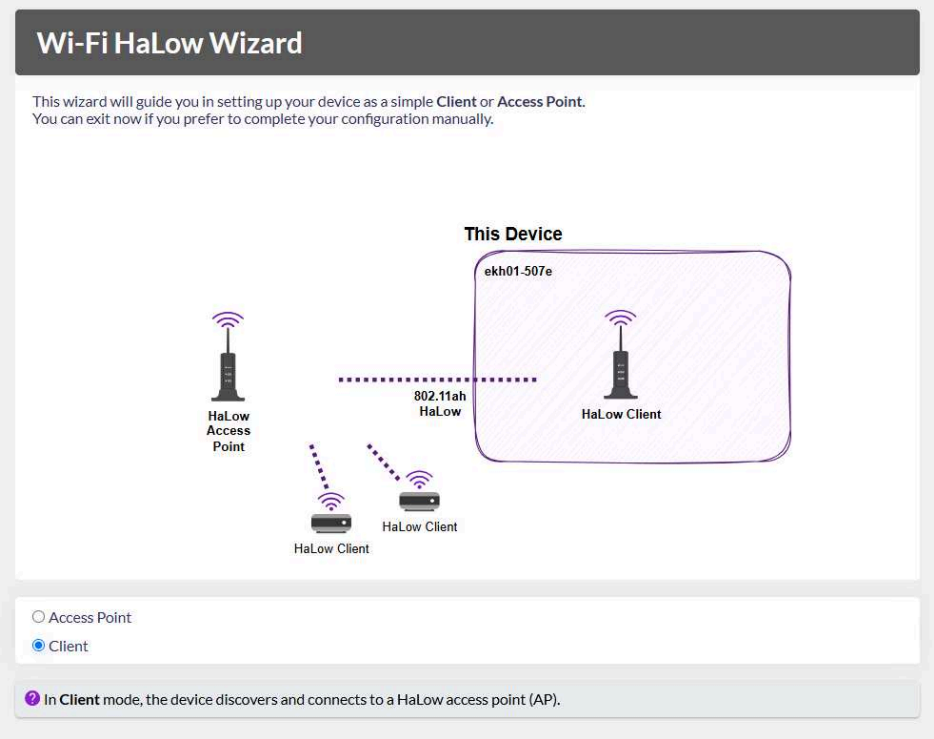


Figure: Standalone AP and STA (Station selection)

4. On the **Connect to a HaLow Network** page, choose **Manual credentials** and then **Scan** to find the SSID you entered for your AP. Enter the previously set passphrase and click **Next**.
5. For the Traffic Mode, select **Bridge** and click **Next**.
6. If your device has 2.4 / 5 GHz Wi-Fi capability, you will be presented with an Enable Access Point toggle which you can switch off if standard Wi-Fi access is not needed. Click **Next**.
7. Once you have saved the configuration by clicking Apply, you should disconnect your laptop from the Station and reconnect it to the AP. You can find the Station's HaLow IP in the DHCP Leases card on the Home page of the AP web UI.

5.4.3 (Optional) Add Upstream Internet Connectivity

In many situations it is helpful to have an upstream connection to the internet. The following instructions outline how to connect your AP to an upstream gateway which can provide internet access to the HaLow devices.

It assumed that the upstream gateway provides the following:

- a DHCP server to allocate an address to your AP.
- a DNS server, provided via a DHCP offer.
- a gateway address, assigned via a DHCP offer.

The steps are as follows:

1. Connect a laptop to your HaLow device as described in Initial Setup and navigate to the admin interface.
2. If the wizard does not come up because you've already configured your device, go to **Wizards** in the side menu.
3. On the **Upstream Network** page, choose **Ethernet**, and set the **Traffic Mode** to **Router**.
4. Apply the configuration on the final page.
5. Use an Ethernet cable to connect your AP to the existing network.
6. To access the device's admin interface again the default web UI IP address may be used over the HaLow link or the user will need to determine the address allocated the network's DHCP server. See section for how to reset a device if access is lost.

5.4.4 (Optional) Add upstream internet connectivity via LTE

Using an LTE dongle for upstream internet connectivity is also supported (instead of via the Ethernet port). An LTE dongle will be automatically discovered and shown as another option available in the **Upstream Network** page (pictured). Follow the steps from the previous Add Upstream Internet Connectivity section, but choose **Dongle** when selecting the upstream network to use.

Note: Connectivity via the LTE dongle behaves the same as Ethernet.

Note: Only dongles that do not require a specific driver are supported (such as via CDC/RNDIS).

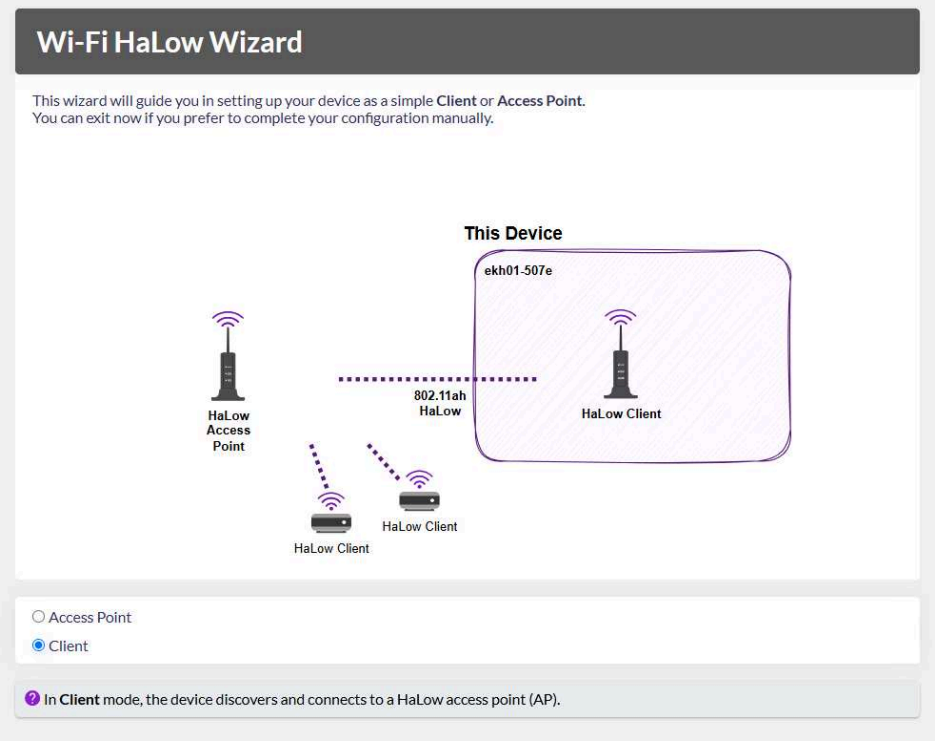


Figure: Upstream Network selection

5.5 Virtual Wire - Layer 2 Bridging

The following outlines how to configure the AP and STA per the scenario defined in Using HaLow as a Virtual Wire (Layer 2 bridge).

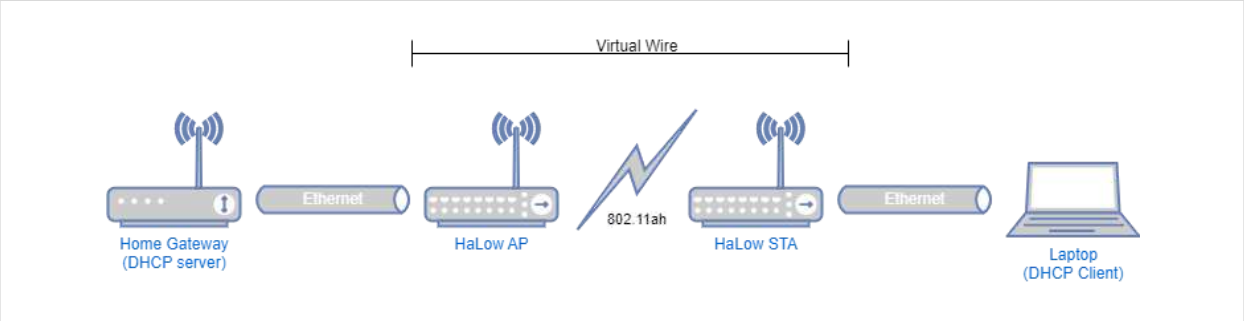


Figure: Using HaLow as a Virtual Wire (Layer 2 bridge)

5.5.1 Access Point Configuration

1. Follow the steps in Initial Setup to connect to the web UI and set the region.
2. For Mode selection, choose **Access Point**
3. Click the **Next** button at the bottom right to go to the next page.
4. On the following pages:
    a. **Setup HaLow Network - AP** has a default SSID and passphrase which you can change if you wish. The Width and Channel options can be left as default. Then click **Next**.
    b. **Upstream Network** should be **Ethernet**. Selecting Ethernet will show a new option for Traffic Mode which should set to **Bridge**. Click **Next**.
5. You can then Apply your configuration on the final page.

To access the device's admin interface again, you will need to check the IP address allocated by your network's DHCP server. See Restoring Factory Settings for how to reset your device if you lose access.

5.5.2 Station Configuration

Follow the same steps as instructed in the Standalone AP and STA Station Configuration section. Once you've saved your configuration, however, your station's IP address will be allocated by your network's DHCP server and will be accessible on that network.

5.6 Non-Standalone AP with Routing

This section outlines how to configure the AP and STA per the scenario defined in Non-Standalone Access Point with Routing.

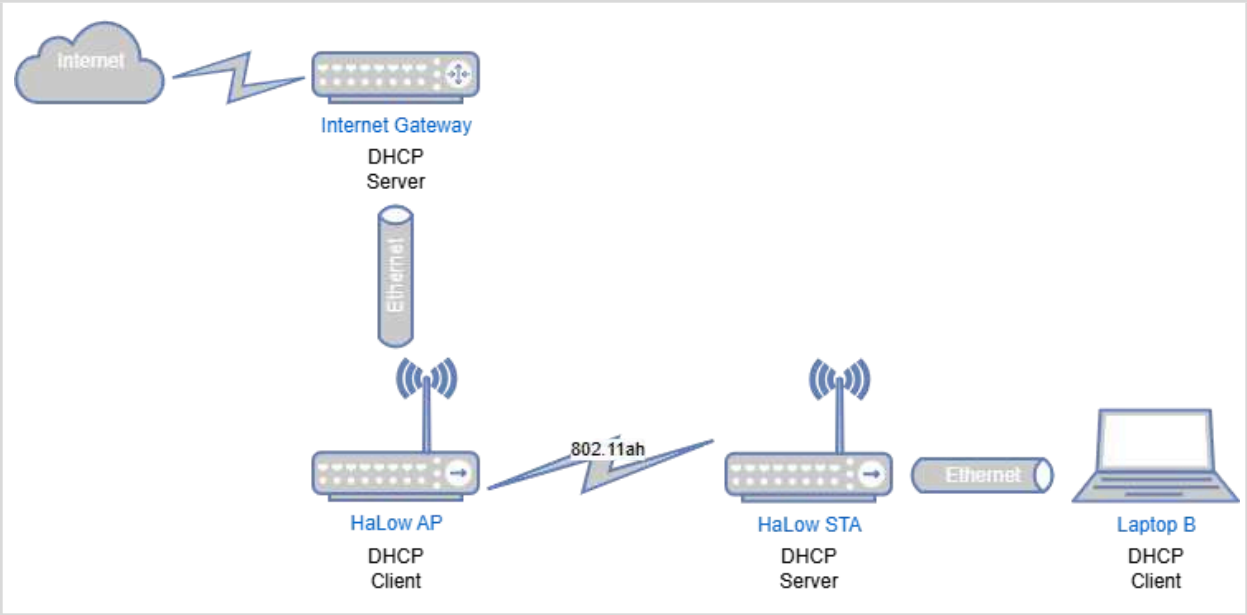


Figure: Non-Standalone AP with Routing

5.6.1 Access Point Configuration

Follow the steps in the Standalone AP and STA Access Point Configuration and Add Upstream Internet Connectivity to configure an Access Point with **Uplink** set to **Ethernet** and **Device Mode** set to **Router**.

5.6.2 Station Configuration

Follow the STA configuration for the scenario in Station Configuration, but for **Device Mode** choose **Extender**.

5.7 Setting a Custom Static IP

By default, devices configured as an AP are reachable at a static IP address on the Ethernet or HaLow interfaces. If the two interfaces are bridged together, either they will operate in DHCP client mode or they will be assigned the default HaLow interface IP address depending on uplink configuration. Clients/Stations will use the static IP address for ethernet and DHCP client for HaLow if the interfaces are separate, and DHCP client mode if bridged.

If you would prefer to use a static IP instead of DHCP client mode or wish to change an existing static IP, go to the Quick Config page in the side menu. Take care to avoid running multiple DHCP servers on a single network.

If you use DHCP on your HaLow AP, this will help it identify the IP/hostname of associated devices on the home page and allow you to access them via their hostname (e.g. https://ekh01-9f62.lan). If you switch your Clients to static IPs, make sure you note them down, otherwise it's easy to lose access.

5.8 Using DPP QR Code

Device provisioning protocol (DPP) provides a simple process to onboard stations into an existing wireless network. Station devices are provisioned by scanning a QR code (generated by the STA) with a "configurator" device, such as a mobile phone, already associated with the network.

5.8.1 On the Access Point

No explicit action is required to enable DPP in AP mode. Simply set your device to work as an Access Point with SAE security.

Note that if you are using 802.11s mesh, this does not by default include AP functionality, and the hostapd process will not be started. The hostapd process is required for DPP to function. It is possible to run 802.11s Mesh with an AP, this mode is known as a 'Mesh Gate' - see the Mesh Gate configuration section for details.

The credentials for the AP DPP configurator are set in `/etc/dppd/auth_secrets.txt`, you will be asked for these in the mobile app after selecting the AP to provision a device to. The default username is **morse** and the default password is **HaLow**.

5.8.2 On the Station

To enable DPP through the web UI, after setting your Device as a client via the Wizard, select DPP on the **Connect to a HaLow Network** page where it asks for credentials. Using this approach, a QR page will be shown on the last page of the wizard:

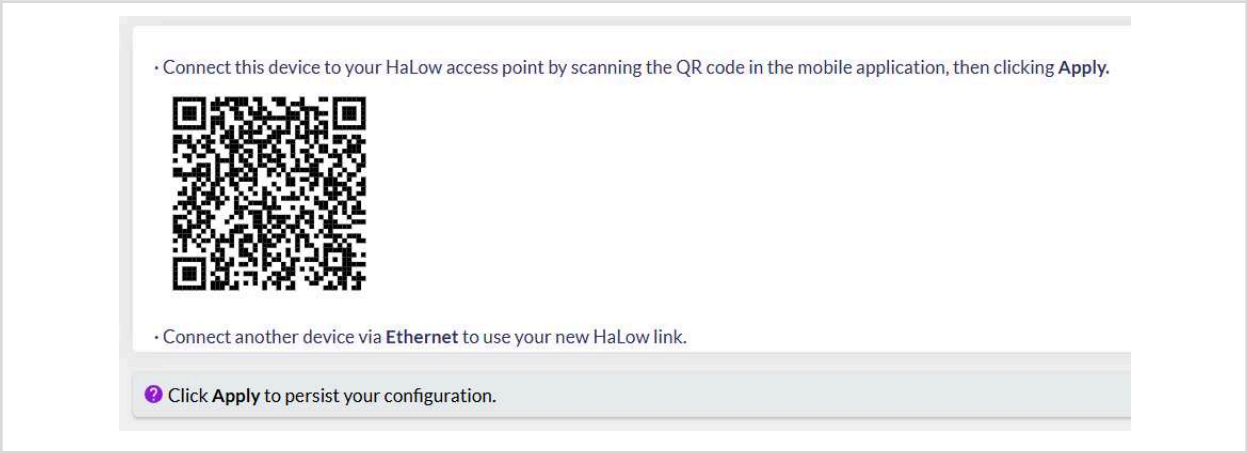


Figure: DPP QR code shown on the last page of the Wizard

Alternatively, you can also view the QR code from the Home page by clicking on the Uplink (HaLow) card and enabling the **DPP QR Code** toggle.

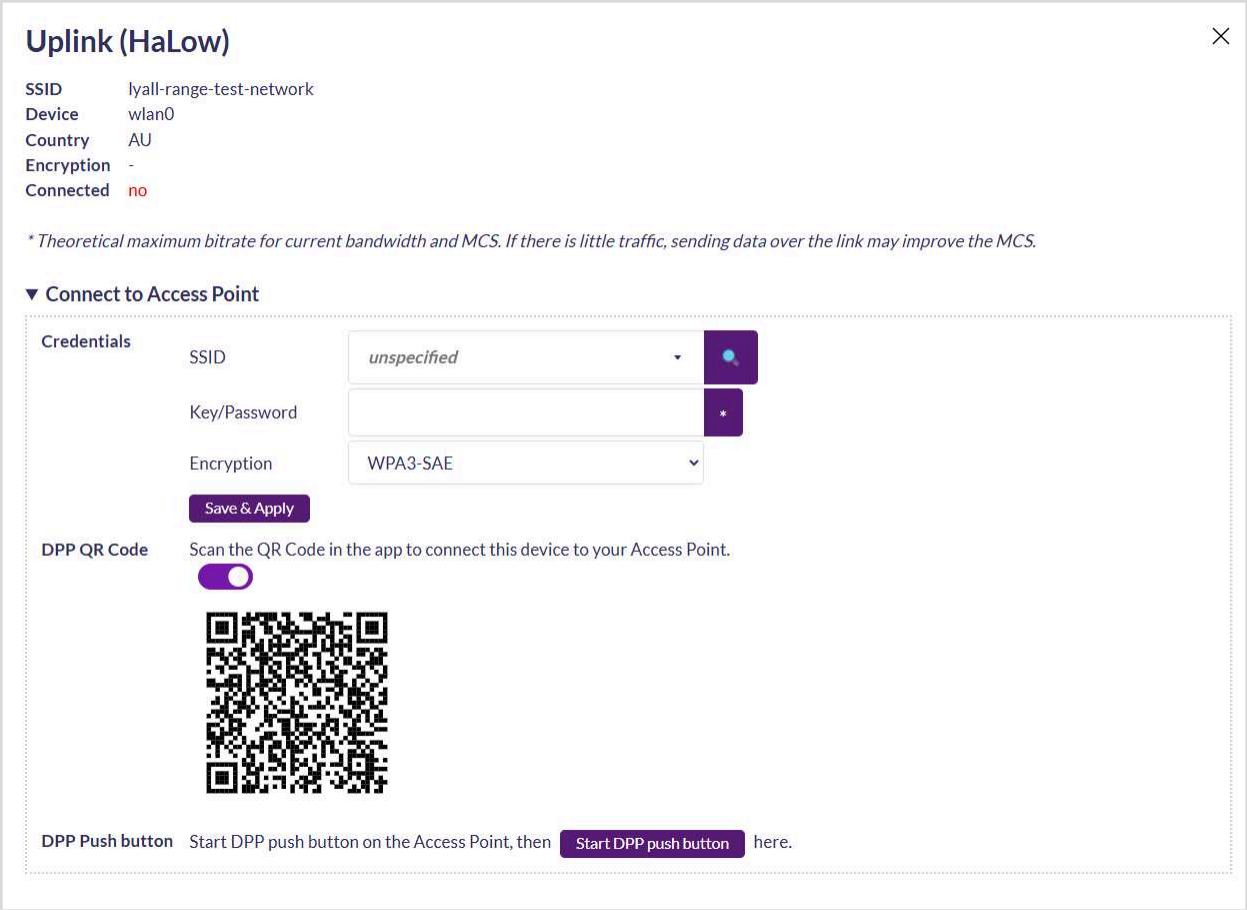


Figure: Enabling the DPP QR Code via the Home page Uplink (HaLow) card

To start provisioning, use the Morse Micro DPP app on the phone to scan the QR code. After a successful provision, SSID, key, and encryption will be set automatically.

Note: The DPP QR code is not persistent on EKH01 devices and will change if updating the image without using the ***Keep settings and retain current configuration*** *option or if updating the image on the microSD card using a computer.*

5.9 Using DPP Push Button

To utilize DPP (Device Provisioning Protocol) using the push button, simply set your device as an Access Point or Station and save the configurations. Then expand the **Access Point (HaLow)** card on the Home page of the Access Point and click on the **Start DPP Push button**.



Figure: Start DPP Push button

Simultaneously on the Station, expand the **Uplink (HaLow)** card on the Home page and click **Start DPP Push button**. Upon successful completion of the DPP process the Station connection status will change to yes.



Figure: The Uplink (HaLow) card showing a successful connection

On the EKH03 the physical reset button on the device can also be used for DPP push button. Pushing the button for less than 2 seconds will start the DPP push button process if the device has been configured as an Access Point or Station. The RGB status LED will indicate that the DPP push button process is running, see the EKH03 LED behavior section for details.

5.10 11s Mesh

802.11s mesh networks aim to increase coverage and range by establishing peer-to-peer links between the various neighbor mesh STAs in the mesh topology. Only mesh capable devices can join the mesh BSS (MBSS) or make use of the mesh functionality provided by the MBSS. Interaction with non-mesh capable devices is handled via mesh gateways (potentially co-located with a non-mesh AP).

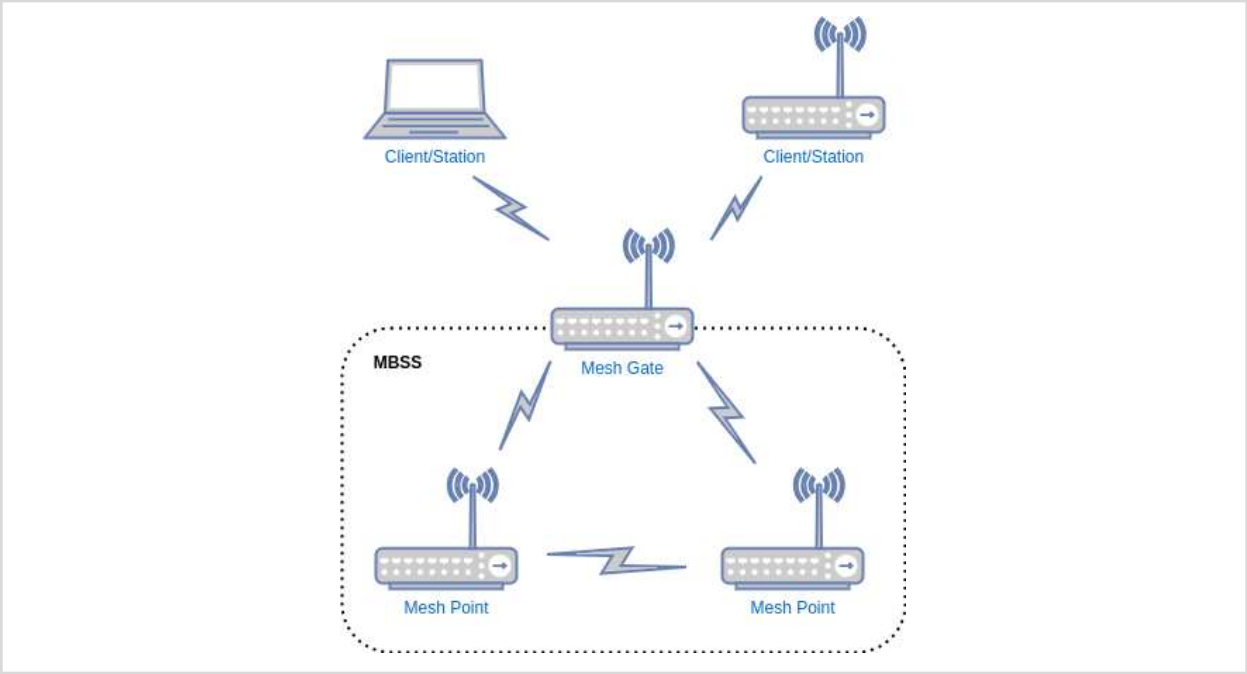


Figure: 802.11s mesh

5.10.1 Mesh Station / Mesh Point Configuration

1. As a prerequisite, ensure that your device is configured with the appropriate region and a channel. Refer to steps in Initial Setup to connect to the device and set the region.
2. Once you decide to configure your device as a **Mesh Point**, navigate to the Wizard selection page in the side menu and select the **802.11s Mesh** Wizard.
3. As a first step, choose **Mesh Point** as the mode and then click **Next**.

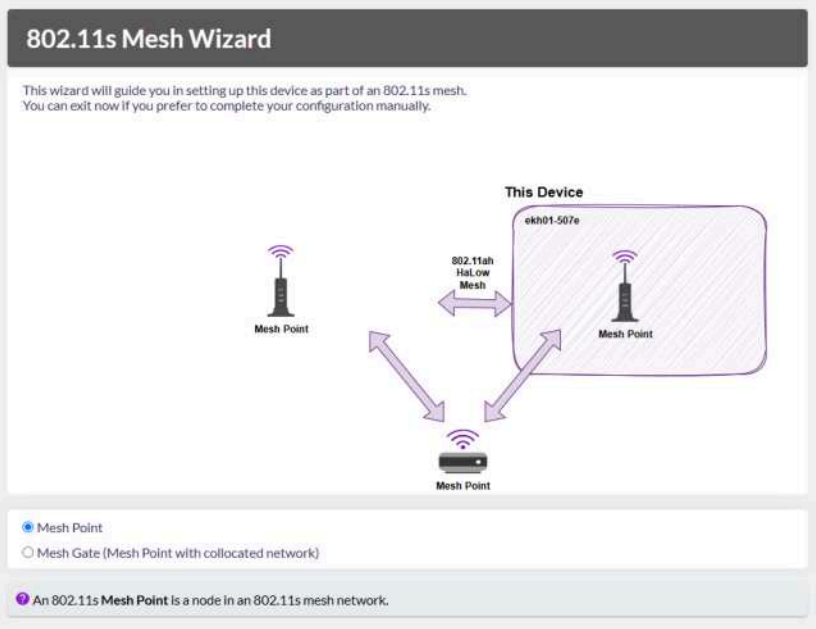


Figure: 802.11s Mesh Wizard (Point selection)

4. On the **Setup Mesh Network** page set an appropriate Mesh ID, encryption and passphrase. Then click **Next**.
5. For the **Traffic Mode**, select **Bridge** and click **Next**.
6. After saving the configuration (by clicking **Apply**), disconnect your laptop from the Mesh Point and connect it either to the Mesh Gate or another Mesh Point to integrate it into the Mesh Network and obtain an IP address for the device. To access the Mesh Point's admin interface again, navigate to the Home page of the Mesh Gate's admin interface and inspect the **DHCP Leases** card.

5.10.2 Mesh Gate Configuration

1. As a prerequisite, ensure that your device is configured with the appropriate region and a channel. Refer to steps in Initial Setup to connect to the device and set the region.
2. Once you decide to configure your device as a **Mesh Gate**, navigate to the Wizards config in the side menu and select **802.11s Mesh Wizard**.
3. As a first step, choose **Mesh Gate** as the mode and then click **Next**.

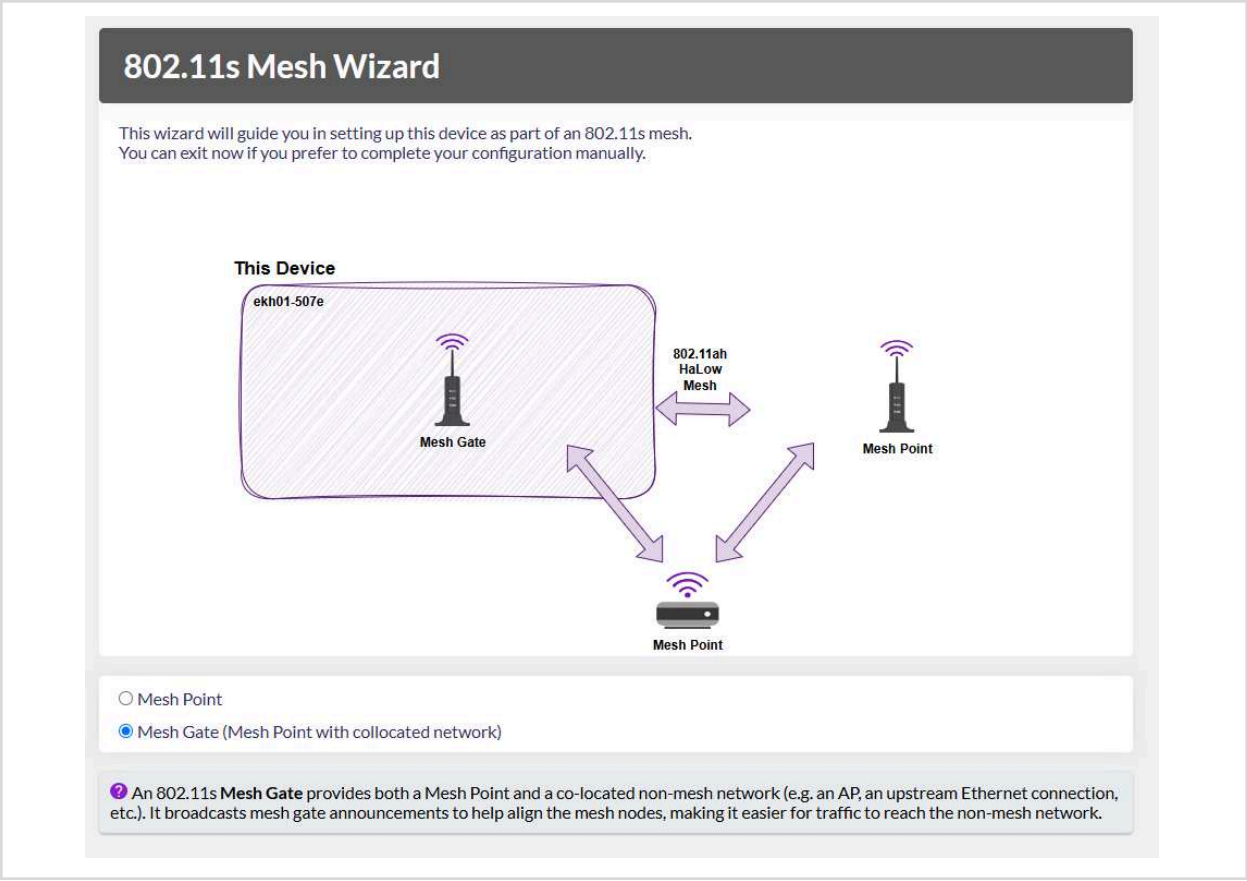


Figure: 802.11s Mesh Wizard (Gate selection)

4. On the **Setup Mesh Network** page set an appropriate Mesh ID, encryption and passphrase. Then click **Next**.
5. On the following page, **Upstream Network** should be **None**. Click **Next**.
6. For a Mesh Gate, you have the option to set up an additional **HaLow Wi-Fi Access Point** interface (Co-located AP) alongside the Mesh interface to extend the HaLow network if needed. On the **HaLow Wi-Fi Access Point** page, if you enable the AP, then ensure to fill in the SSID, encryption and password for that interface. Please note, that this AP interface that is created is always bridged with the Mesh interface in the Mesh Gate mode. Then click **Next**.

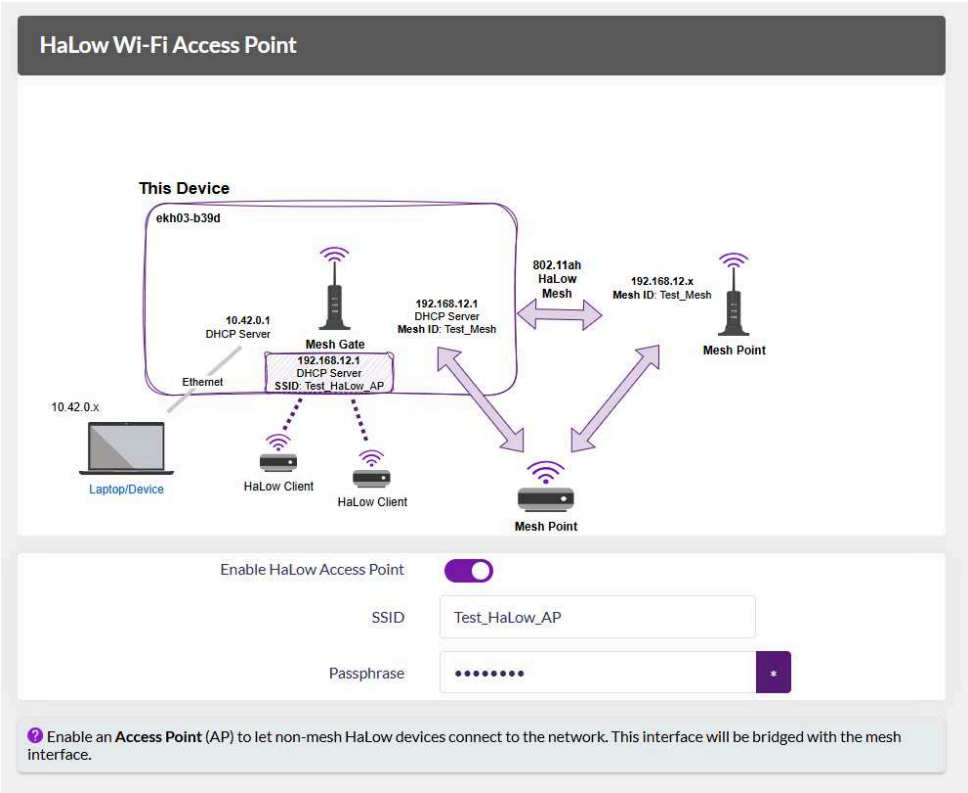


Figure: 802.11s Mesh Wizard (Co-located Wi-Fi HaLow Access Point configuration)

7. You can then **Apply** your configuration on the final page.

5.10.3 (Optional) Add Upstream Internet Connectivity in Mesh Gate Mode

Typically a Mesh Gate is a device that provides access to one or more distribution systems via the wireless medium for the mesh basic service set (MBSS). In many situations it is helpful to have an upstream connection to the internet. The following instructions outline how to connect your Mesh Gate to an upstream internet gateway.

It assumed that the upstream gateway provides the following:

- a DHCP server to allocate an address to your AP.
- a DNS server, provided via a DHCP offer.
- a gateway address, assigned via a DHCP offer.

The steps are as follows:

1. Connect a laptop to your HaLow device as described in Section 3.1 and navigate to the admin interface.
2. Go to the **Wizard** config in the side menu and select **802.11s Mesh Wizard**.
3. Repeat the steps 3 and 4 as in Mesh Gate Configuration.

4. On the **Upstream Network** page, choose **Ethernet**, and set the **Traffic Mode** to **Router**.
5. Refer to step 6 in Mesh Gate Configuration and then proceed to **Apply** the configuration on the final page.
6. Use an Ethernet cable to connect your Mesh Gate to your existing network.
7. To access the device's admin interface again, you can access the default IP address over the HaLow link or you will need to determine the address allocated by your network's DHCP server. See Restoring Factory Settings for how to reset your device if you lose access.

5.10.4 Additional 802.11s Mesh Settings

In addition to configuring the 802.11s Mesh settings through the wizard, you can access further options by navigating to the Wireless page under the Network sub-menu in the side bar. Click **Edit** next to the Mesh Interface to access and adjust the advanced mesh settings available in this section.

If you decide to enable B.A.T.M.A.N for the Mesh Interface in Mesh Gate mode, remember to include the AP interface (if it exists) in the same network. You can achieve this by adding the same network name to the AP interface in **Network** > **Wireless** > **Edit (AP)** > **Interface Configuration** > **Network**.

To configure a static IP address see Setting a Custom Static IP.

5.10.5 11s Mesh Visualizer

You can use the **802.11s Mesh visualizer** by clicking on the **802.11s Mesh Topology** card on the Home page of any 802.11s mesh devices, which will display the topology as seen by that node.

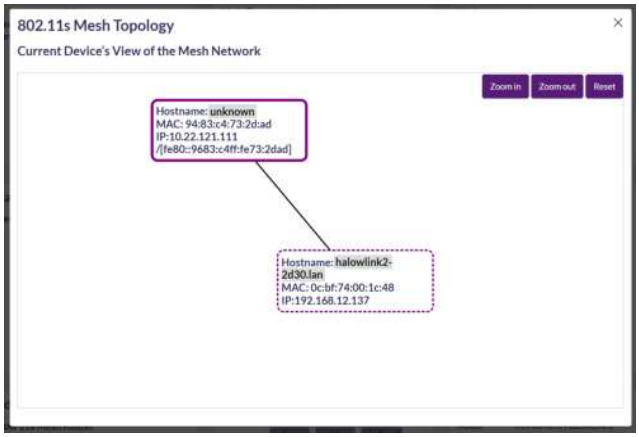


Figure: 802.11s Mesh visualizer

5.11 EasyMesh

EasyMesh is a Wi-Fi branded, standards-based solution for meshing together access points to provide an extended coverage area (but with reduced bandwidth available to stations). EasyMesh forms a tree structure with a controller at the root that controls the mesh network, and agent APs that connect both upstream towards the controller and downstream towards stations. Stations are agnostic to mesh, and continue to connect to the closest AP as usual.

The current implementation supports up to 4 agents in addition to the controller, with at most 2 agents between the controller and a station.

5.11.1 EasyMesh Controller Configuration

1. As a prerequisite, ensure that your device is configured with the appropriate region and a channel. Refer to steps in Initial Setup to connect to the device and set the region.
2. Once you decide to configure your device as an EasyMesh Controller, navigate to the **Wizard** config in the side menu and select **EasyMesh Wizard**.
3. As a first step, choose **Easymesh Controller** as the mode and then click **Next**.

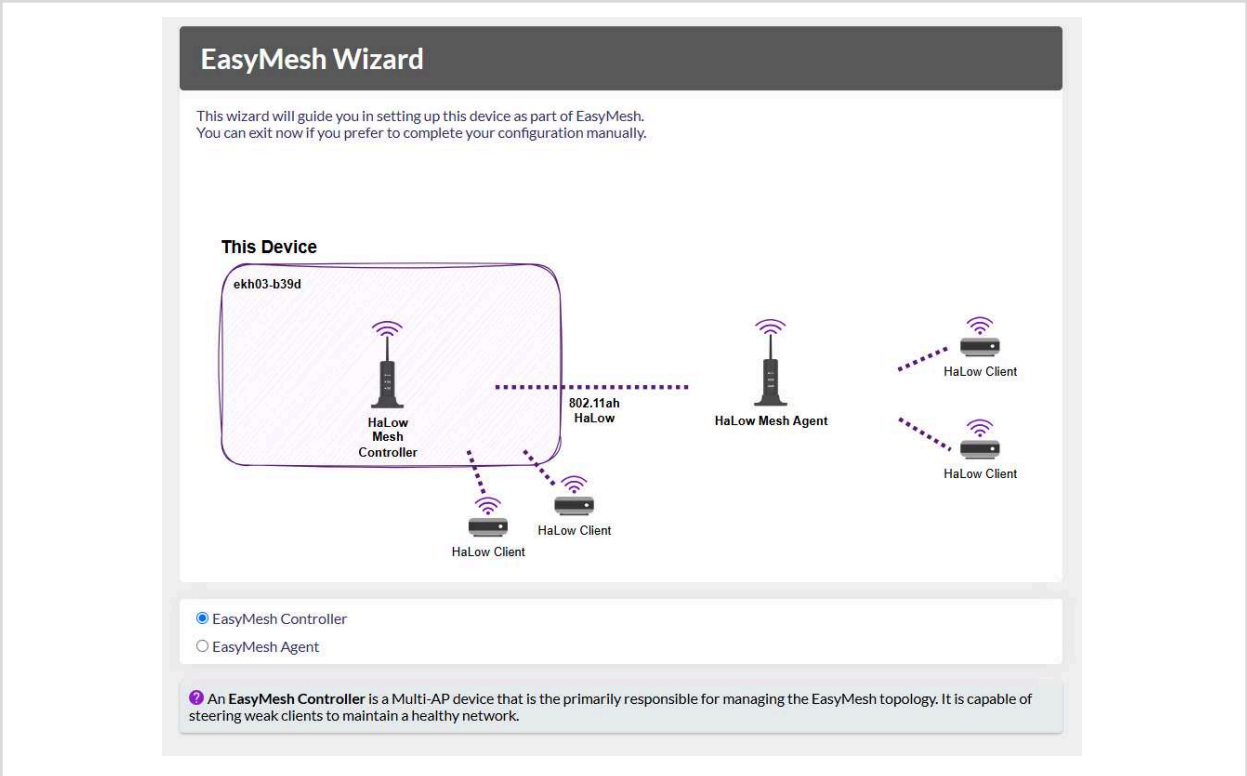


Figure: EasyMesh Wizard (Controller selection)

4. On the **Setup EasyMesh Network** page set an appropriate SSID and passphrase. Encryption is defaulted to WPA3-SAE. Then click **Next**.
5. On the following page, **Upstream Network** should be **None**. Click **Next**.
6. You can then **Apply** your configuration on the final page.
7. Proceed to Pairing EasyMesh Devices for connecting with other Agents.

5.11.2 EasyMesh Agent Configuration

1. As a prerequisite, ensure that your device is configured with the appropriate region and a channel. Refer to steps in Initial Setup to connect to the device and set the region.
2. Once you decide to configure your device as an EasyMesh Agent, navigate to the **Wizards** config in the side menu and select **EasyMesh Wizard**.
3. As a first step, choose **EasyMesh Agent** as the mode and then click **Next**.

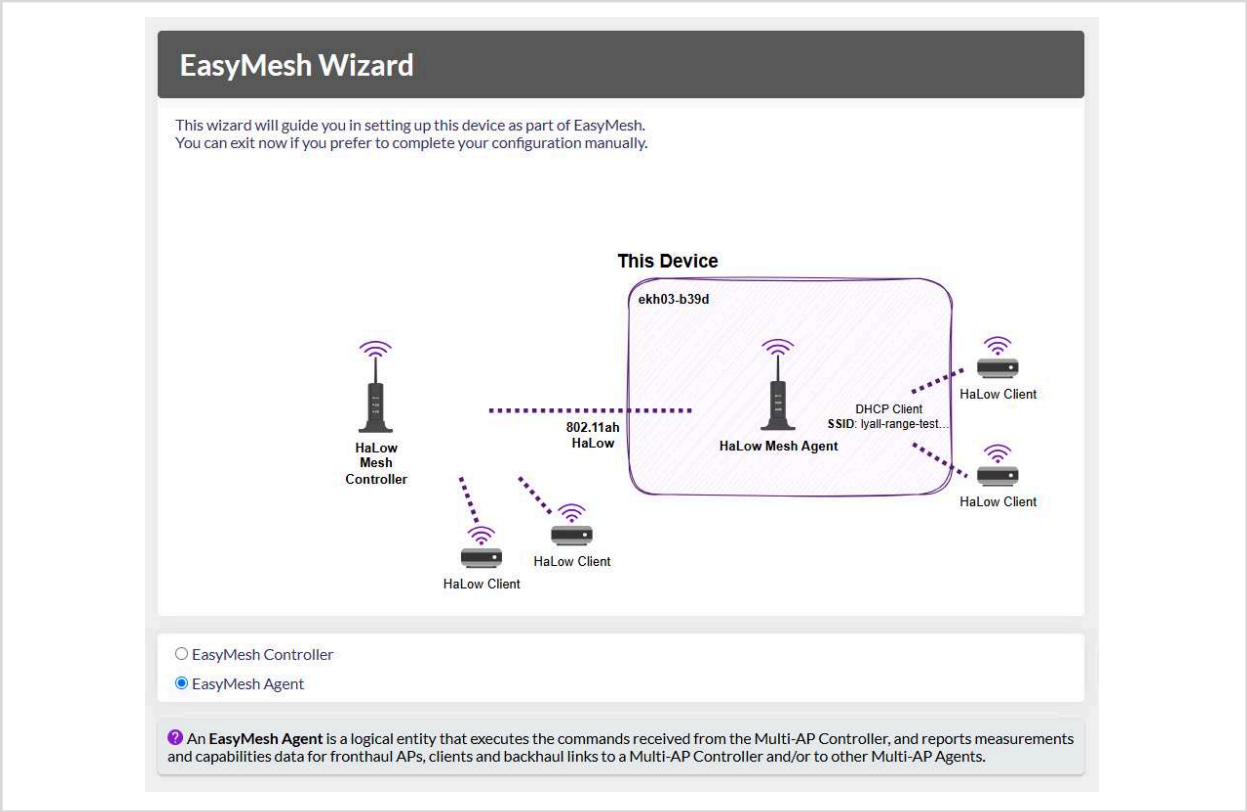


Figure: EasyMesh Wizard (Agent selection)

4. For the Traffic Mode, select 'None' and click Next.
5. Proceed to section Pairing EasyMesh Devices for connecting with an EasyMesh Controller or other Agents.

5.11.3 Pairing EasyMesh Devices

1. As a prerequisite ensure that your device is configured either as an EasyMesh Controller or Agent following in the steps in EasyMesh Controller Configuration or EasyMesh Agent Configuration respectively.
2. On the EasyMesh Controller, Navigate to the Home page in the UI, and once the Agent Status shows **Active** in the EasyMesh card, click on the **Start WPS** button.
3. Simultaneously, on the EasyMesh Agent device, navigate to the Home page in the UI, and in the EasyMesh card, click on the Start WPS (client) button to pair itself with the Controller.
4. Upon successful completion of WPS pairing the Home page will show the **Agent Status** as **Active**.

5.11.4 EasyMesh Visualizer

To confirm that EasyMesh has been enabled and is working, status information is available on the **Home** page of the **EasyMesh Controller**, which displays the number of **Connected Agents**. Clicking on **Connected Agents** in the EasyMesh card on the UI will open a topology diagram:

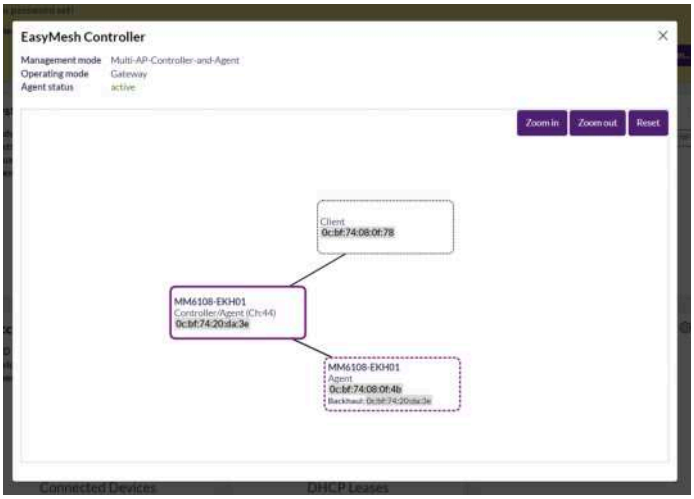


Figure: EasyMesh Wizard (Agent selection)

Logs are also available from **Status** > **System Log** when EasyMesh is enabled. If these are not visible, you may need to logout of the frontend due to caching.

5.12 Monitor Mode

It is possible to configure the HaLow interfaces to operate in monitor mode, which allows the device to capture all 802.11ah packets on the air within the specified channel and bandwidth. To enable monitor mode make the following changes on the Quick Config page:

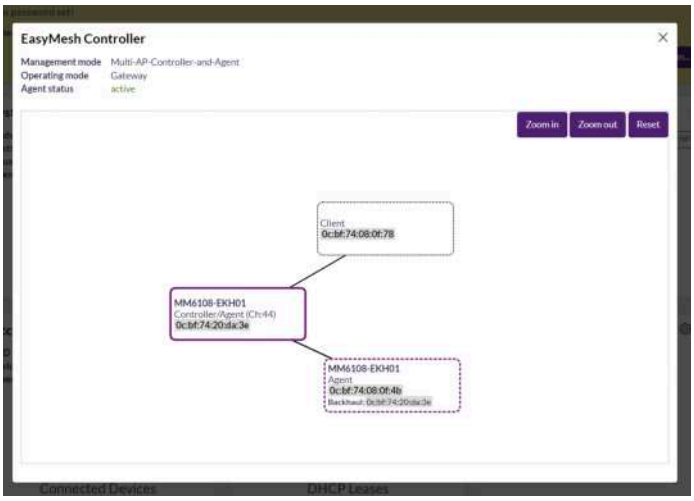


Figure: Quick Config page HaLow Wireless Interface Monitor Mode

Monitor mode can also be configured on the **Network** > **Wireless** page by selecting the interface that you want to change to monitor mode and manipulating the popup dialog. This page provides additional options such as frequency, bandwidth, primary channel bandwidth and primary channel index. These additional options are not available in the Quick Config page.

After setting the interface into monitor mode, it is possible to start capturing packets using the **morse0** interface (not the interface that is set to monitor mode). Further changes to monitoring parameters can be made using morse_cli command, for example:

```
morse_cli -i wlan0 channel -c 908000 -o 8 -p 2 -n 0
```

Where the full syntax for this command is as follows:

```
morse_cli -i wlan0 channel [-aj] [-c <freq>] [-o <operating BW>]
[-p <primary BW>] [-n <primary chan index>]
```

Note: Please keep in mind that parameter changes are done via the monitor interface (wlan0, wlan1, ...) but the packet dump is done through morse0 interface.

5.13 Native S1G (beta)

*Note: This section **ONLY** applies for MM8108 devices. Additionally, native S1G is a beta feature for experimental use, but not intended for production use yet.*

As S1G technology grows increasingly more and more mature, it needs to be made available in the upstream Linux kernel to accelerate both user adoption and development. Morse Micro is committed to the open source effort, but this is not going to be an overnight feature-complete driver. As such, the initial driver submitted to the kernel will be minimal and incrementally built upon.

To switch from using the default Morse Micro driver to the native S1G driver, a button is present in the **Wireless** section of the **Quick Config** page.

Note that this is a destructive process — your existing configurations will be cleared.

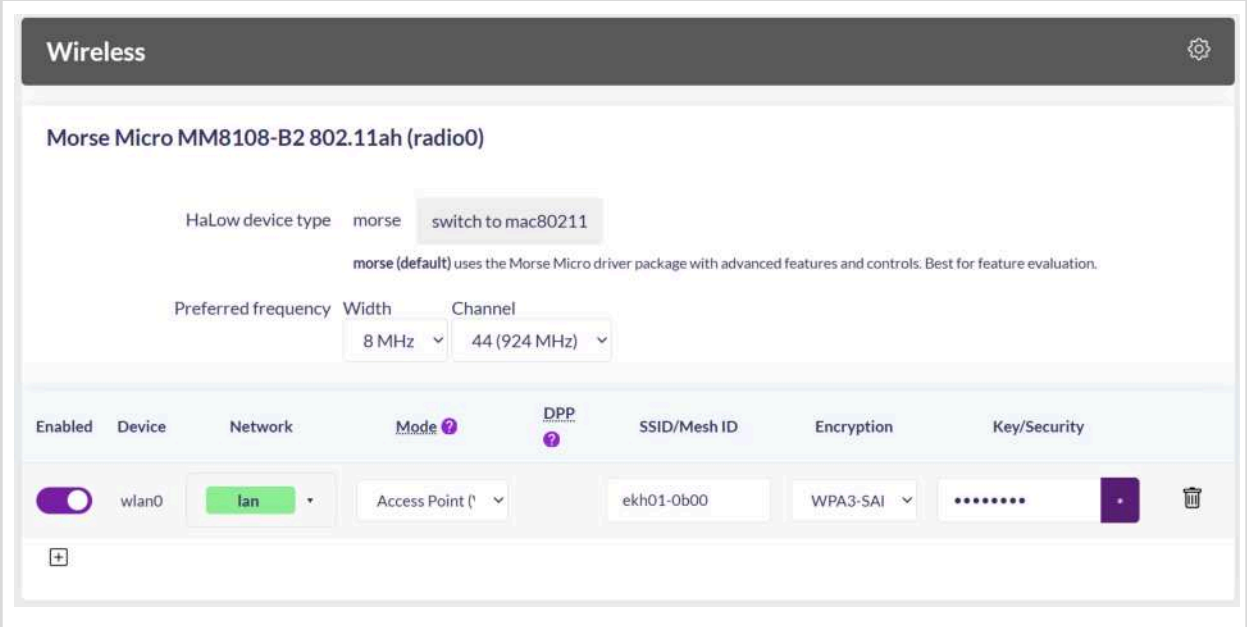


Figure: The driver switch button under Wireless on the Quick Config page.

Note: Native S1G is currently only available for the US and does not support mesh capabilities.

6 Performance Testing and Evaluation

6.1 Wavemon and Ping Testing

Wavemon provides a powerful way to quickly test the performance and quality of a HaLow connection in the field. All that is required is a mobile phone, HaLow AP, HaLow STA (with a suitable power supply) and a USB-Ethernet cable to connect the mobile to one of the HaLow devices. The diagram below shows how to setup the equipment:

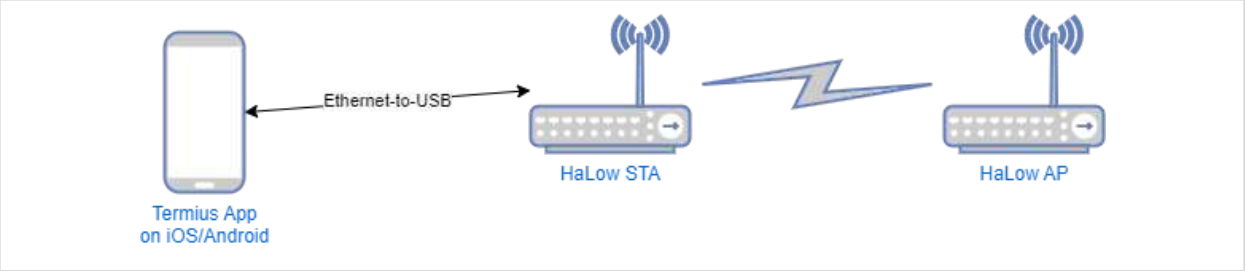


Figure: Wavemon and Ping Testing

To run wavemon, the mobile device will need to be able to run an SSH session (e.g. using Termius for Android/iOS). Once an SSH session has been started run the command pt from the command line interface and a wavemon ping test will begin:

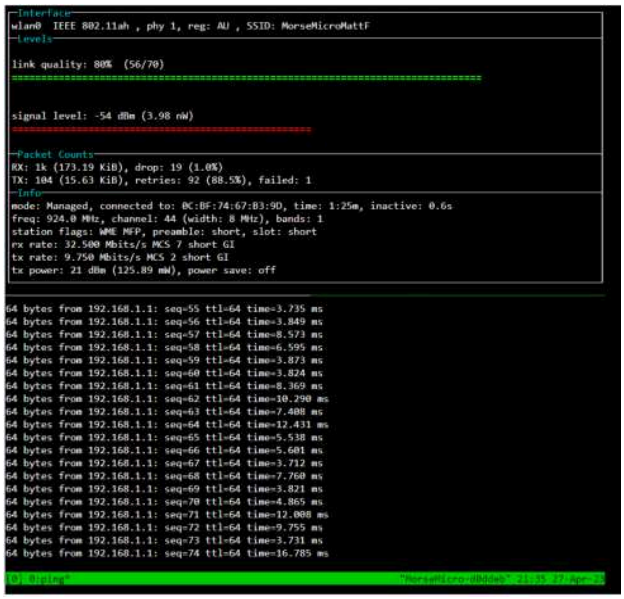


Figure: Wavemon CLI output

Note that `pt` will ping 192.168.12.1 by default, but an alternate address can be provided as an argument to the script, e.g. `pt 1.2.3.4`.

6.2 Setting Up iPerf3 Traffic Testing

iPerf3 testing provides a tool for analysing the quality of HaLow connections by sending a stream of traffic and measuring the speed, throughput and latency.

To run an iPerf3 test you must have at least 2 devices:

- One device must be selected to act as the **iPerf3 server**.
- One device must be selected to act as the **iPerf3 client**.
- The devices must be associated via HaLow. The devices can be connected in any of the network configurations previously mentioned in Configuration of Operating Modes like Standalone AP/STA, mesh or otherwise.
- The **IP address** of the iPerf3 server must be known.
- The CLI or UI of both devices must be accessible.

6.2.1 Via the CLI

To run an iPerf3 test, you need at least two associated devices where the command line is accessible via the terminal or SSH. The devices can be connected in any of the network configurations previously mentioned in Configuration of Operating modes like Standalone AP/STA, mesh or otherwise.

Ensuring that you have access to the CLI of both devices (either via SSH or the Terminal page), login to the device which you have selected as the **server** and run:

```
iperf3 -s
```

Once the server has successfully started you should see something like “Server listening on 5201 (test #1)” appear on the screen.

Now, log into the CLI of your device which you have selected as the **client** and run:

```
iperf3 -c <server_device_ip>
```

This will start sending a stream of randomly generated TCP traffic over the HaLow connection to the server. The test should run for 10 seconds by default and provide you with a report at the end:

```
[ ID] Interval           Transfer     Bitrate
[  5]   0.00-10.09  sec  19.2 MBytes  16.0 Mbits/sec
```

There are many different flags and options that can be set to fine tune these tests. The full list can be seen by running `iperf3 --help`.

6.2.2 Via UI

You can also run iPerf3 in server and client mode via the web UI on the **Diagnostics** page under the **Network** sub-menu.

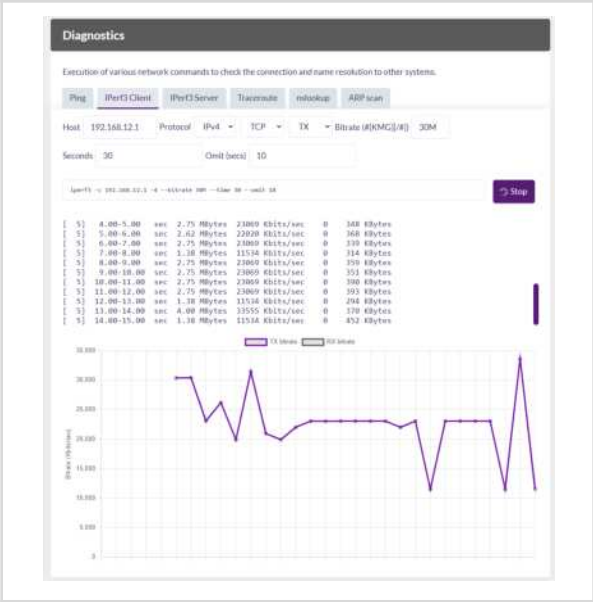


Figure: iPerf3 client on the Diagnostics page

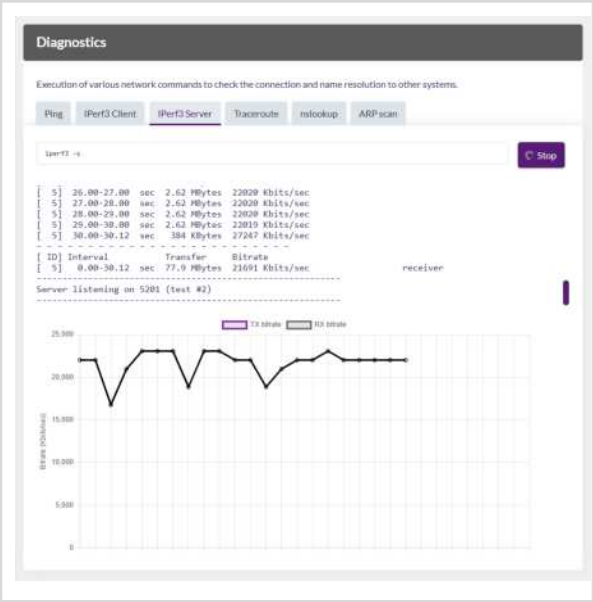


Figure: iPerf3 server via the Diagnostics page

6.3 Range Testing

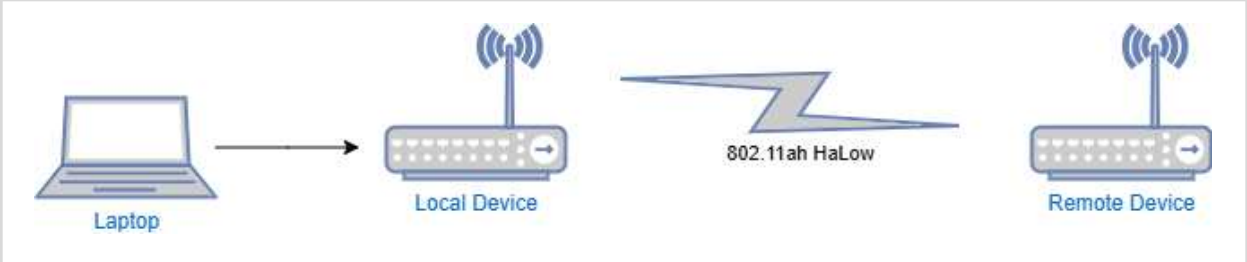


Figure: Range Test device setup

The Range Test application is designed as a simple way to analyse HaLow network performance by automating `iperf3` tests and collecting real-time statistics. It is a useful tool for quickly assessing signal strength, data throughput, and connection quality across different environments.

Note: ensure that all devices under test are running the same version of Morse Micro OpenWrt.

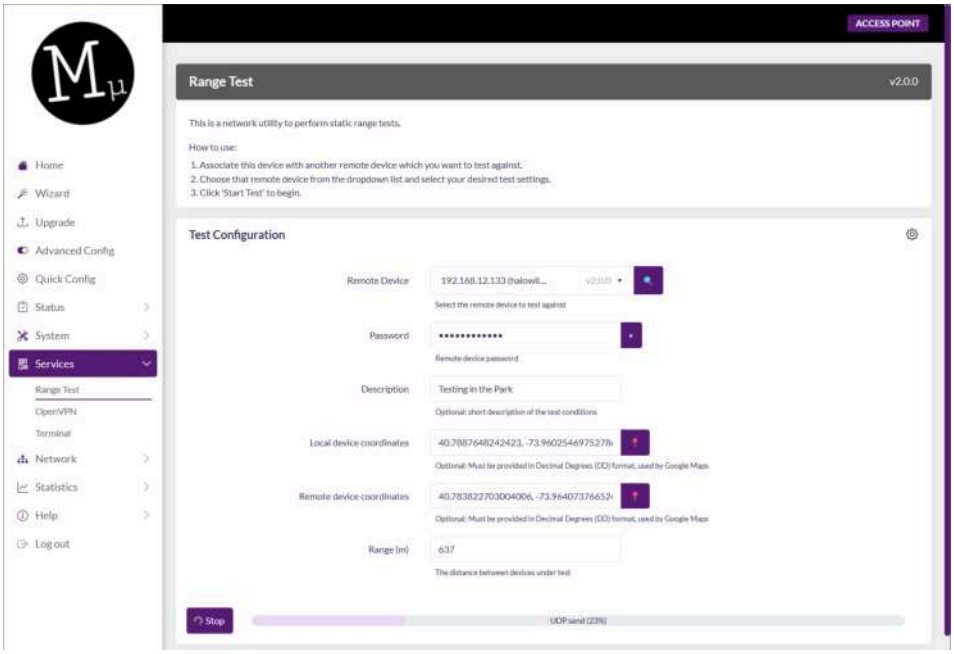


Figure: Range Test setup

To use the range testing tool:

1. Connect the devices being tested on the same HaLow network of any type outlined in this guide (Default AP/STA configurations are the most reliable).
2. Set up the devices in the desired locations and select a local device to connect a laptop to.
3. Select the IPv4 address from the **Remote Device** dropdown. If it does not appear on first load, click the 🔍icon to run another discovery. If this fails to find another device, you may enter a known IPv4 address into the **custom** field pictured.

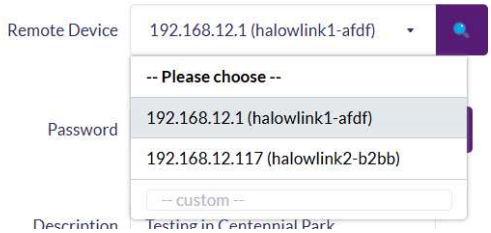


Figure: The Range Test setup Remote Device dropdown

4. Fill in the **Password** field using the remote device password (not the Wi-Fi password).
5. (Optional) Provide user notes about the test in the **Description** field.
6. (Optional) If you want to log the coordinates of the local and remote devices and have the range calculated automatically enter the decimal degree values into the local and remote device coordinates fields respectively. Right-click a location in Google Maps to copy its coordinates to your clipboard in the correct format.

7. Click **Start Test** and a progress bar will appear showing the status of the current test. This can be cancelled at any time by clicking the **Stop** button.

After the test completes, a row in the **Results Summary** subsection should appear, as pictured below. This result can be deleted by clicking the **trash icon**, or a JSON file containing all the raw data can be downloaded via the **Download** button, which is intended to help engineers with remote debugging. The **Download Results Summary (CSV)** button yields a CSV file of all the data represented in the results summary.

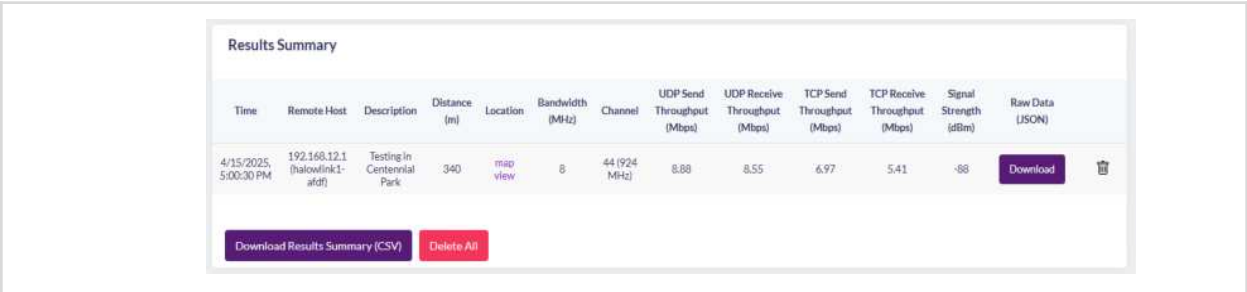


Figure: The Range Test results

Note: test results are volatile to avoid overwhelming limited memory resources and will be deleted if power is removed or the device is rebooted.

Warning: if the throughput numbers are returning ~40 Mbps for UDP and 900+ Mbps for TCP then the test is likely defaulting to run over an Ethernet connection. This should be avoided.

By clicking the **cog icon** in the top left corner of the **Test Configuration** subsection, the data directions, test length and protocols which will be tested can be modified.

The remote device will always act as the `iperf3` server and use the same command:

```
iperf3 -s -1 --json
```

The local device will always act as the `iperf3` client, running a command in the following format:

```
iperf3 -c <ip> <protocol> <direction> -t <time> -O <omit> --json
```

- **Protocol**: For UDP tests the protocol arguments become `-u -b 40M/30` whereas for TCP tests only `-Z` is set.
- **Direction**: For Send tests no arguments are set whilst Receive tests set the `-R` flag.
- **Time & Omit**: If Quick Tests are selected in the Advanced Configuration dialog, these arguments will become `-t 10 -O 2`. By default the tests will run for longer with `-t 70 -O 10` to allow 10 seconds for rate control to settle and get 60 seconds worth of usable data.

6.4 Video Streaming

OpenWrt supports video streaming functionality where cameras attached to stations can stream video to an AP, viewable via the UI. This functionality includes automatic discovery of cameras on the HaLow network which are running compatible firmware (detailed below in station configuration). Any ONVIF-compliant cameras on the network which support H.264 streams will be detected automatically.

Warning: due to resource constraints this feature is only currently available on the EKH01 and EKH19.

6.4.1 Accessing the Video Streams

In the web UI navigate to the **Cameras** page under the **Services** sub-menu in the side bar.

The AP will automatically discover all ONVIF cameras on the network. Note that it will only scan the selected network attached to the interface listed next to the Discover button. After scanning it will automatically start streaming from the discovered cameras.

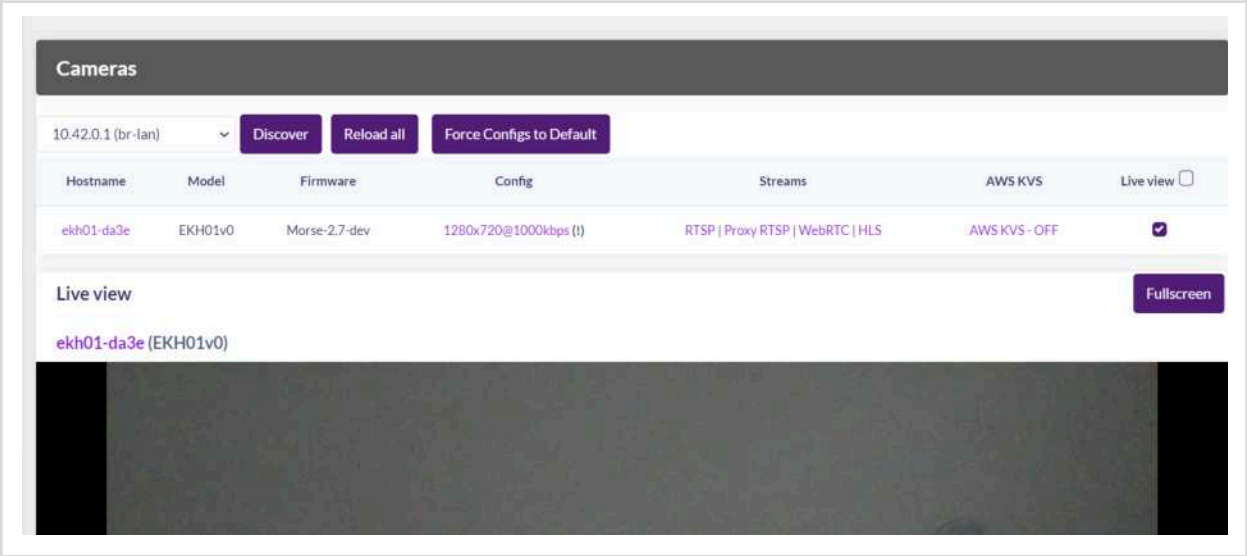


Figure: Cameras page controls

The checkboxes under the **Live view** column are used to select which video streams should be displayed.

6.4.2 Configuration

The following options are available for configuring video streaming:

- **Discover** finds all cameras on the selected interface.
- **Reload All** reloads all the previously discovered camera streams.

- **Force Configs to Default** changes all camera configurations to the default configuration. Hovering over the button displays the default settings.
- **Config** opens a window to modify the camera's configuration.

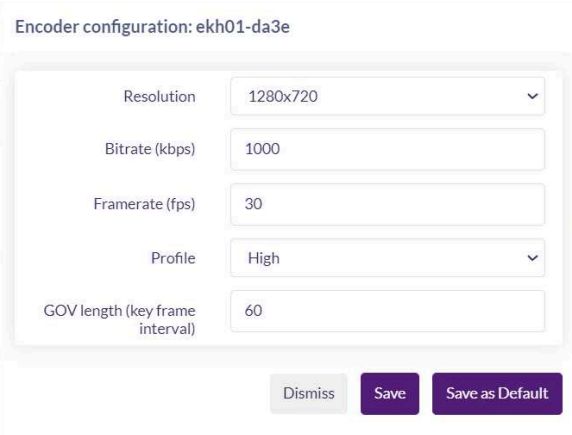


Figure: Stream configuration on the Cameras page

    - **Resolution** of the camera
    - **Bitrate** of the video stream
    - **Framerate** of the video stream
    - **Profile** for the H.264 compression
    - **GOV length** sets number of frames between each key frame
- **Streams** selects the type of stream you want to view (opens in a new window).
- **Live View** selects whether to show a live stream on the current page (via WebRTC).
- **Fullscreen** shows a combined fullscreen view of all the currently enabled streams. To see a fullscreen view of an individual stream, hover over the stream to bring up the video controls.



Figure: Fullscreen view of 6 simultaneous camera streams on the Cameras page

6.4.3 Live View

Cameras can also be configured from the **Live view** window, that includes the resolution, bitrate, framerate and brightness.

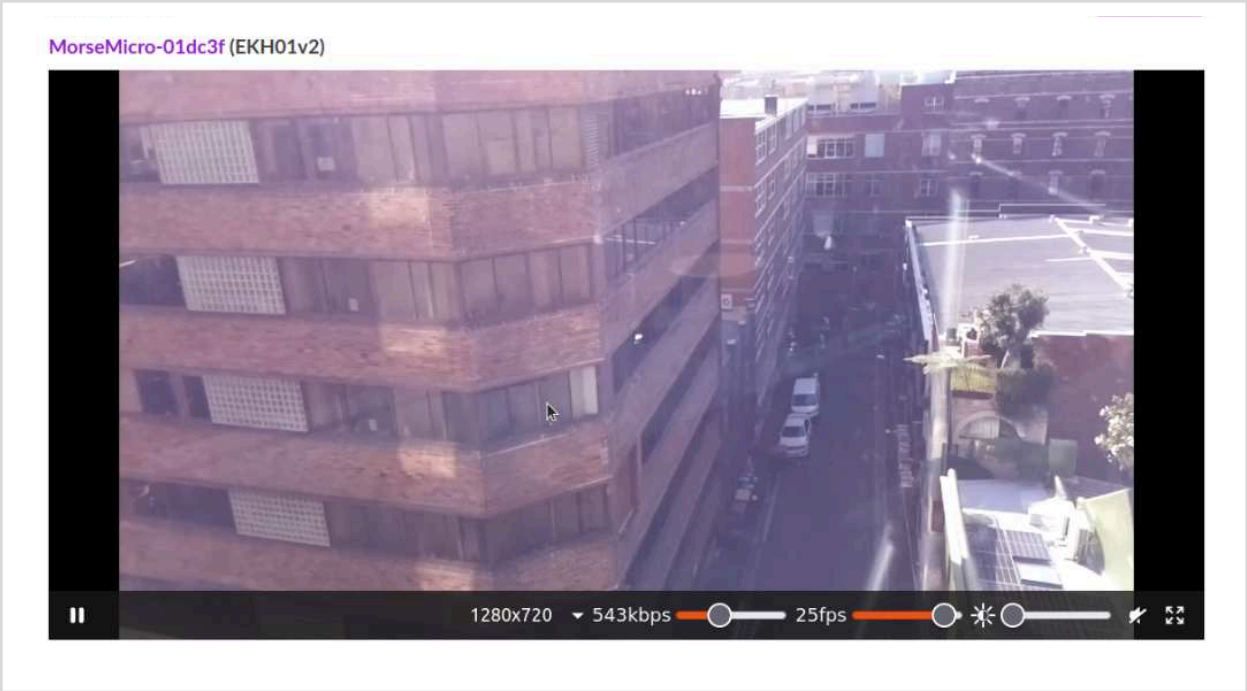


Figure: Cameras page stream example with video controls

6.5 HaLow Diagnostics

The HaLow Diagnostics tool shows live Wi-Fi HaLow link quality and the stability of the radio environment. It is accessed via the **HaLow Diagnostics** page under the **Network** sub-menu and provides two tabs: **Peer Link Quality** and **Link Planner**. Sampling the radio's signal, noise, and interference counters over a moving time window, it reports whether interference is present and how stable the link is 6.5.1, and the link's quality and achievable data rate 6.5.2.

6.5.1 Link Stability and Interference Characterisation

The tool continuously characterises the radio environment on the operating channel and reports two separate results in the **Interference Live** panel of each tab:

- The interference characteristics present on the channel.
- An overall impact on link stability due to interference.

The tool recognises five interference characteristics, each shown as **Detected** or **Clear**.

Characteristic	What it indicates
Cellular	Very high-power interference from outside the HaLow band, typically from a nearby cellular transmitter.
In-channel	A narrowband interferer is active inside the operating channel.
Close-channel	Strong interference sitting just outside, but close to, the operating channel.
Same-channel	Another Wi-Fi HaLow network operating on the same channel.
Adjacent-channel	A Wi-Fi HaLow network likely present on an adjacent or partly overlapping channel.

Table: Interference Characterisation

The tool predicts link stability as one of three verdicts:

Link Stability Verdict	Meaning
Link is stable	No interference was detected during monitoring.
Interference spikes seen, link still stable	Spikes occurred, but the weakest measured signal stayed comfortably above the level the link needs.
Interference can introduce instabilities	Spikes occurred and the weakest signal fell too close to the level the link needs, so occasional drops or rate reductions may occur.

Table: Link stability verdict

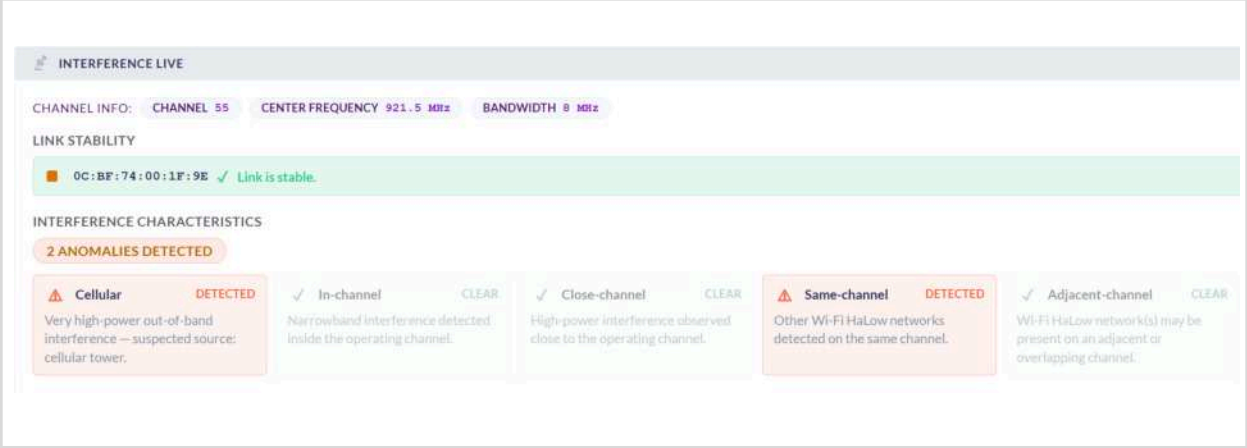


Figure: Live Link Stability and Interference

6.5.2 Link Quality

The quality of a link is rated into five categories that are derived from the predicted Modulation and Coding Scheme (MCS), which in turn is based on the signal-to-noise-and-interference ratio (SINR). A higher SINR yields a better MCS, and therefore a better link quality rating.

Scale	Predicted MCS	Description
Very Poor	None	Conditions are not expected to carry data reliably. The link may fail to establish or drop frequently.
Low	0–1	Only the most robust and slowest rate can work.
Moderate	2–3	The link is usable but limited with lower throughput.
Good	4–6	The link is stable with mid to high throughput.
Excellent	7–9	A strong link with maximum throughput.

Table: Link quality scale

For each window, the tool predicts link quality under three profiles, each answering a different question:

Profile	How it is calculated	Answers
Typical	Most representative signal, noise and interference.	How does the link behave most of the time?

Profile	How it is calculated	Answers
Conservative	A weaker-than-usual signal and a higher-than-usual noise and interference level.	How does it behave during normal rough patches?
Worst case	The weakest signal and highest noise floor observed, including the noise estimated during interference spikes.	What to design for if the link must not drop?

Table: Link quality profiles

The profiles mean the same thing in the **Peer Link Quality** and **Link Planner** tabs. The key difference is that the signal (RSSI) value in **Peer Link Quality** is measured from the real peer, while in **Link Planner** it is predicted by the propagation model where the user supplies either a distance to the far end or a known path loss. The noise and interference are always measured live on the configured channel.

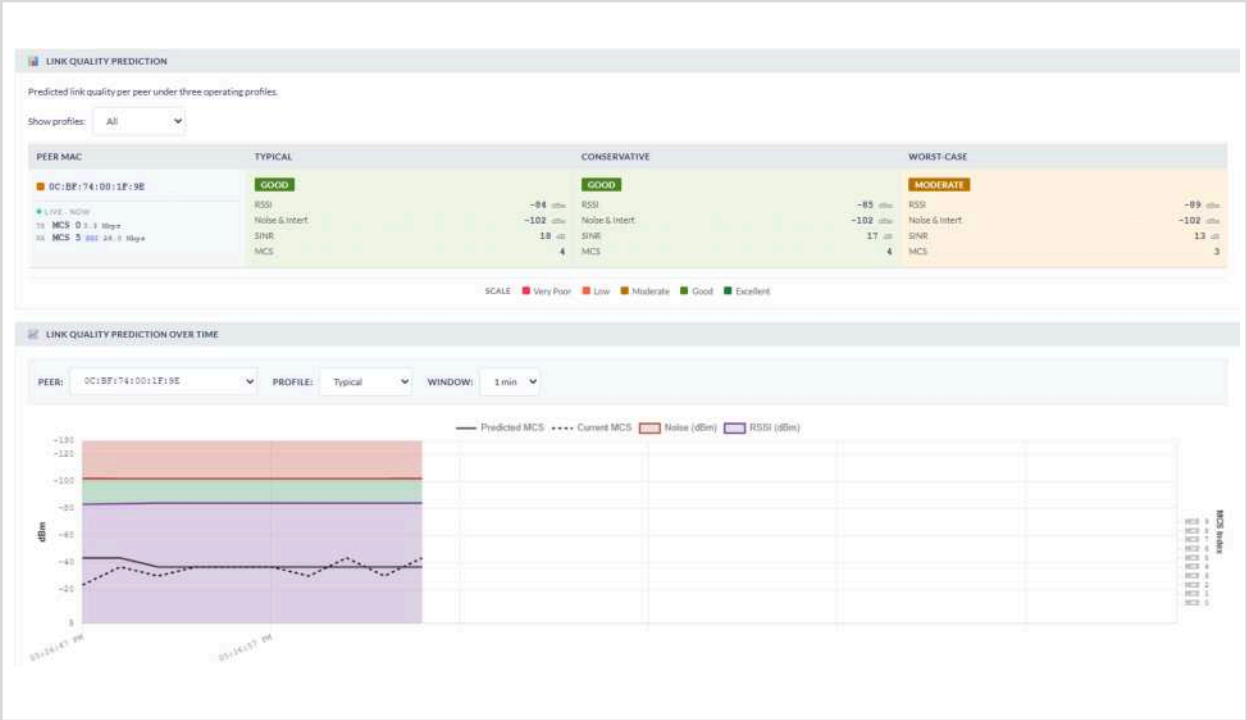


Figure: Link Quality Prediction

Note: a session is kept alive only while the page is actively requesting results. If the page is closed or left inactive for about 15 seconds, the session ends automatically and its resources are released.

Warning: a Link Planner session is intrusive. Starting an estimate places the HaLow firmware into an active medium-evaluation state which may reduce throughput and affect the performance of peer links. Peer Link Quality and Link Planner are mutually exclusive — starting a planner estimate will stop any Peer Link Quality monitoring already running.

7 Page Descriptions

This section describes some of the pages available in the web UI.

7.1 Home

The Home page provides a comprehensive view of the device's current status. It includes several key sections to help users understand the overall state of their device.

Card	Description
System	Provides general details about the device, including the software version and model name. This section is essential for identifying the device and understanding its current firmware status.
Mode	Displays the operational mode of the device and includes a configuration representation with a topology diagram. This visual aid helps users see how their device is configured and operating within the network.
HaLow Network Status	Shows the current status of the HaLow network, including any connected clients if the device is functioning as an Access Point. This information is crucial for monitoring the HaLow network's health and connected devices.
Uplink Information	Provides the status of the uplink connection, offering details about connectivity. This section ensures users are aware of their device's connection to the wider network or internet.
Local Network Info	Gives detailed information about the local network, including DHCP lease information if the DHCP server is enabled. This section helps users manage their local network and understand IP address assignments within their network.
Network Interfaces Information	Displays the groupings of network interfaces, showing how they are organized and connected. This information is important for managing and troubleshooting network interfaces.

Table: Home page card descriptions

Note: clicking on the ⚙ *symbol in each of these cards will navigate to the corresponding advanced configuration page.*

7.2 Upgrade

To update your software, use your browser to access the web interface as described in Initial Setup. Then select **Advanced Config**, and you should see the **Upgrade** page:

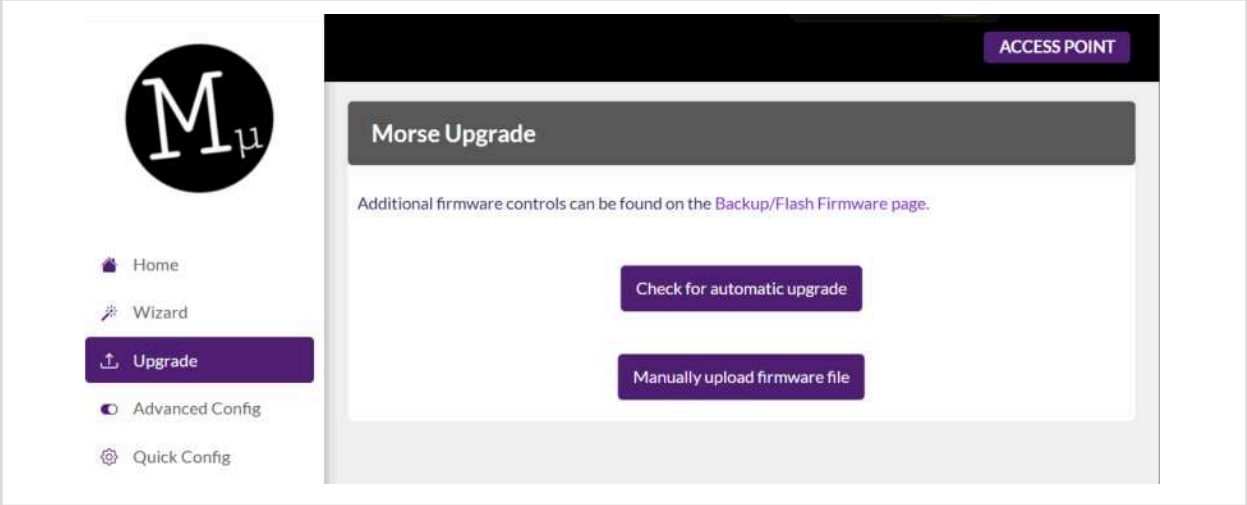


Figure: The Upgrade page

- The **Check for automatic upgrade** button will attempt to obtain the new version of firmware from the Morse Micro servers. This requires an available internet connection — either through the browser you are using to access the GUI or through the device itself via an uplink or otherwise .
- The **Manually upload firmware file** button will let you upload any compatible firmware.

Note: if your device has internet access, update notifications will appear automatically on the Home page within the Version card.

7.3 Quick Config

For most users the Wizard is sufficient for configuring devices. We recommend using the Wizard for larger changes, such as moving from an Access Point to a Client.

However, if you want to make simple minor changes to your configuration - such as changing your Wi-Fi password or encryption method, or setting a Static IP - you can do so via the **Quick Config** page. To help you understand the changes you're making, these will be reflected in the diagram at the top of the page as you make them.

Warning: it is easy to change the configuration of your device here in a way that causes you to lose access to it, particularly if you're changing network interfaces. If this happens, see the section on Restoring Factory Settings.

For more information about how your device is configured, refer to UI Configuration Architecture.

7.3.1 Network Interfaces

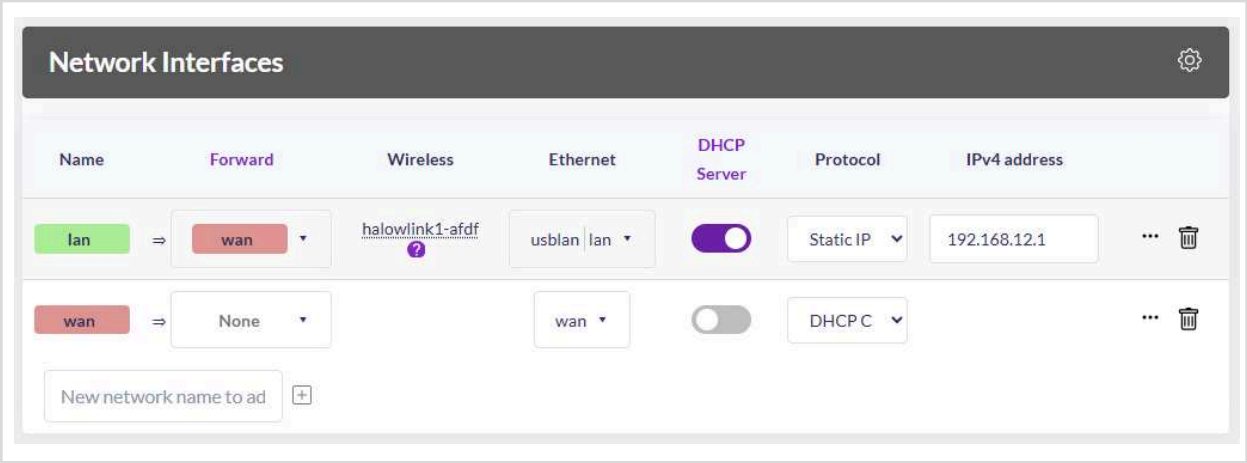


Figure: The Network Interfaces subsection on the Quick Config page.

This section lists the logical interfaces available on your router, each of which can be configured with either a Static IP (and potentially a DHCP Server) or as a DHCP Client. This section allows you to make changes to the Network configuration. The following fields are available:

- **Name** displays the network name. The configurations for this network and the list of Ethernet and Wireless interfaces mapped to this network are displayed in the corresponding row.
- **Forward** enables traffic forwarding between the source and the selected network. Click 'Forward' for advanced firewall settings.
- **Ethernet** selects an Ethernet port and maps it to the specified network.
- **DHCP Server** enables the DHCP server for the network. Click 'DHCP Server' for advanced DHCP and DNS configurations.
- **Protocol** sets either 'Static IP' for manual IP address configuration or 'DHCP client' to retrieve an IP from a DHCP server if available. Using DHCP is generally preferred.

Click the three dots at the end of the row to configure the following options:

- **Netmask** sets the netmask for the interface. This option is available only if the protocol is set to **Static IP**.
- **Gateway** sets the IP address of the upstream gateway for default IP traffic routing. This option is available only if the protocol is set to 'Static IP'.
- Click on the ⚙ symbol, to navigate to the advanced Network Configuration page.

7.3.2 Wireless Section

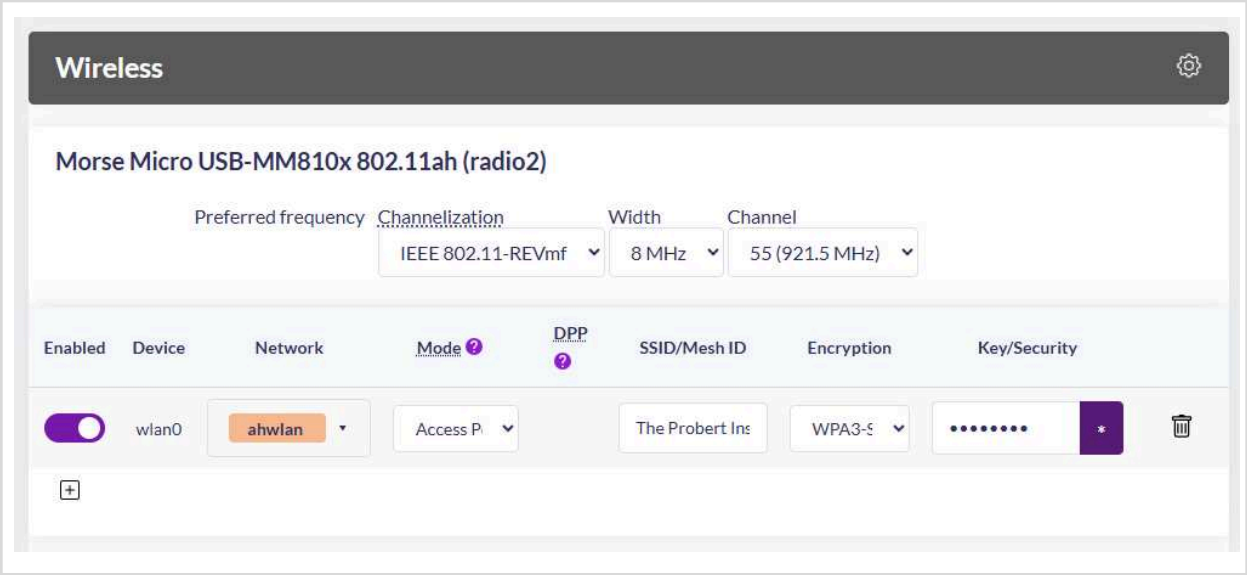


Figure: The Wireless section on the Quick Config page.

This will allow you to configure the Wi-Fi interfaces on your device. The available fields are:

- **Channelization** (AU only) chooses the different channelization schemes available. To view the different channels available with each scheme, visit the Regulatory page.
- **Preferred Frequency** Width selects the operating bandwidth for the HaLow network. The dropdown menu is automatically populated based on the Country selection.
- **Channel** chooses the frequency channel for the HaLow network. The dropdown menu is automatically populated based on the selected Preferred Frequency Width.

For each available wireless interface row there are the following configurations:

- **Enabled** indicates the status of the wireless interface. Toggle the slider to enable or disable the wireless interface.
- **Device** displays the name of the wireless interface.
- **Mode** selects the mode of operation for the wireless interface. Options include: Access Point (WDS), Access Point (no WDS), Client (WDS), Client (no WDS), Mesh Point, Ad-Hoc (IBSS), Monitor and None.
- **DPP** lets the device broadcast a DPP preference for connection establishment.
- **SSID/Mesh ID** configures the SSID for connection. The field initially shows the currently configured SSID. In Client mode, clicking the 🔍 icon will scan for visible HaLow networks and populate the dropdown with visible SSIDs. If the SSID is not visible, you can manually enter the name and press enter to set it. In Mesh Point mode, enter the Mesh ID here.
- **Encryption** selects the encryption method for data sent over the HaLow network. Available methods include OWE, SAE, Enterprise security (EAP), and None (Open security):
    - OWE (Opportunistic Wireless Encryption) - Ensures privacy between the station and access point without requiring a password or station authentication.

    - SAE (Simultaneous Authentication of Equals) - Uses pre-shared passwords for symmetric encryption, suitable for mesh networks.
    - EAP (Enterprise Security) - Requires RADIUS server configuration for the Access Point and TLS, TTLS, or PEAP authentication credentials for the Client.
- **Password** is visible when SAE is selected as the encryption method. It configures the password used to authenticate and set up encryption between the station and the access point.

The Wireless section of the Quick Config page also provides support for Multiple Basic Service Set Identifiers (MBSSID), allowing users to run multiple virtual interfaces on a single physical device.

Additional interfaces can be added by clicking the ⊞ button at the bottom of the section. Each new entry will create a separate BSSID which will appear as a new row in the table above sharing the same radio settings of the device but allowing for customization of SSID, encryption, network and key. This is useful for setting up distinct logical interfaces on the same hardware.

Click on the ⚙ icon to navigate to the advanced Wireless Configuration page and click **Edit** on the network's corresponding row to modify more advanced network settings such as DTIM Interval, Station Inactivity Limit, Beacon Interval and more.

7.3.3 Advanced Usage

The **Quick Config** page is designed to correspond with the underlying text based configuration. You can use the **cog icons** on the **Quick Config** to access these pages, which will allow you more flexibility in your setup. It will also allow you to simply **Save** rather than **Save & Apply**, which will let you view and apply or revert the proposed changes by clicking on the **Unsaved Changes** indicator at the top right.

For more about the underlying configuration format (known as UCI), see UI Configuration Architecture.

7.4 Status

7.4.1 Realtime Graphs

This page displays live HaLow statistics graphs, showing the last 3 minutes of data, updated at a 3 second interval. On the Wireless tab (pictured below) the graphs are signal strength, data rates, and MCS respectively.

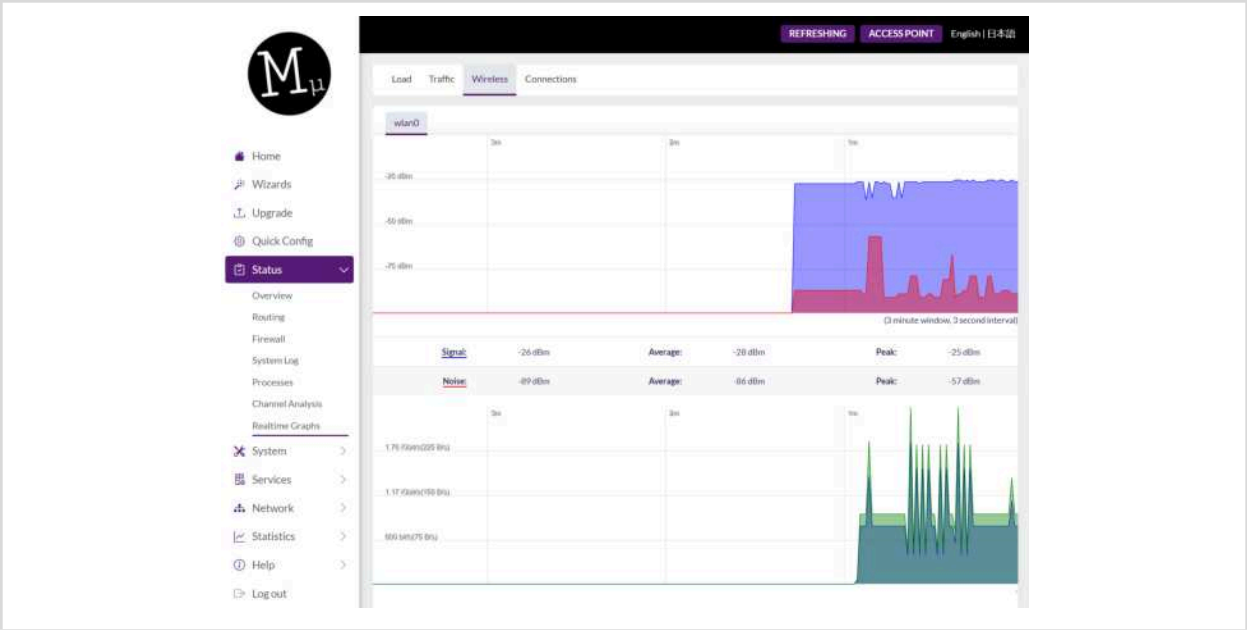


Figure: Wireless tab on the Realtime Graphs page

7.5 System

7.5.1 Administration

On this page you can enable SSH and turn off HTTPS enforcement.

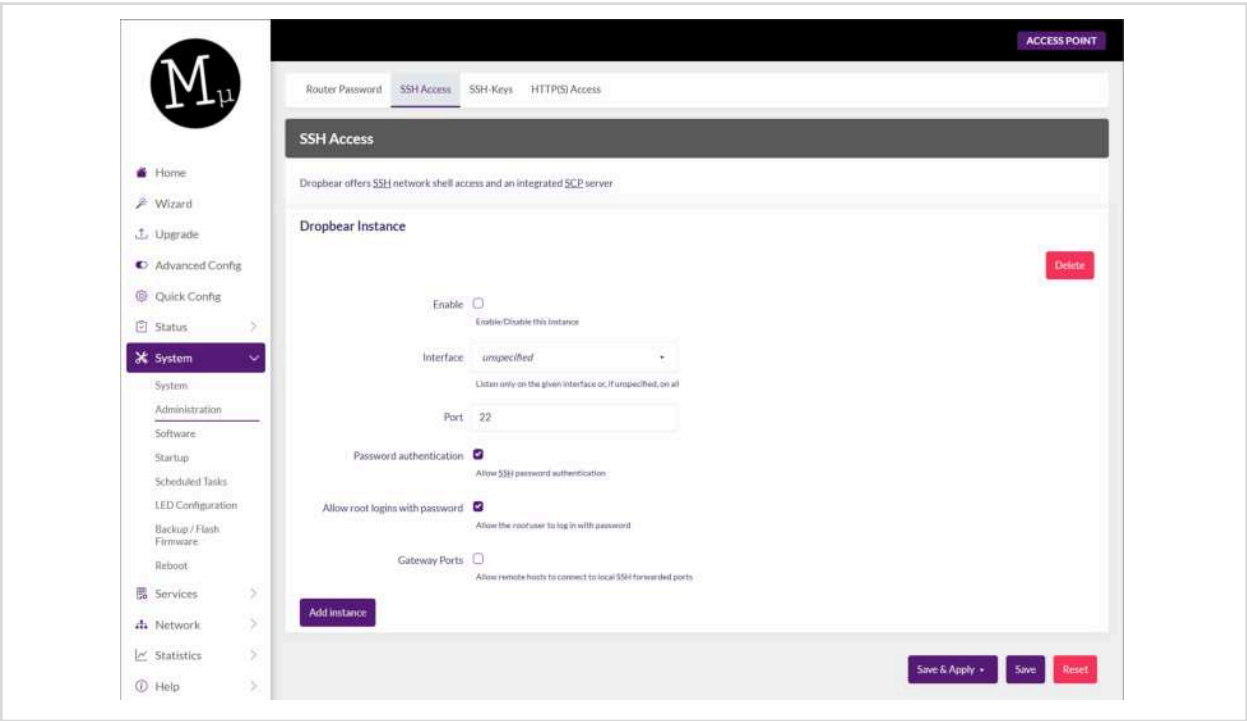


Figure: The SSH Access tab on the administration page

7.5.2 Backup / Flash Firmware

The **Perform reset** button allows you to factory reset the device.

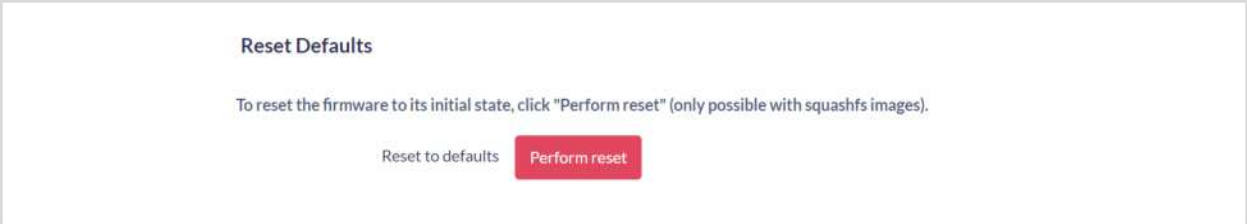


Figure: The Perform reset button on the Backup / Flash Firmware page

7.6 Services

7.6.1 Terminal

For easy access to the Linux console, you can start a terminal on the device by going to Terminal page under the service sub-menu. Note that you will have to re-enter your device password (refer to the sticker on your device).

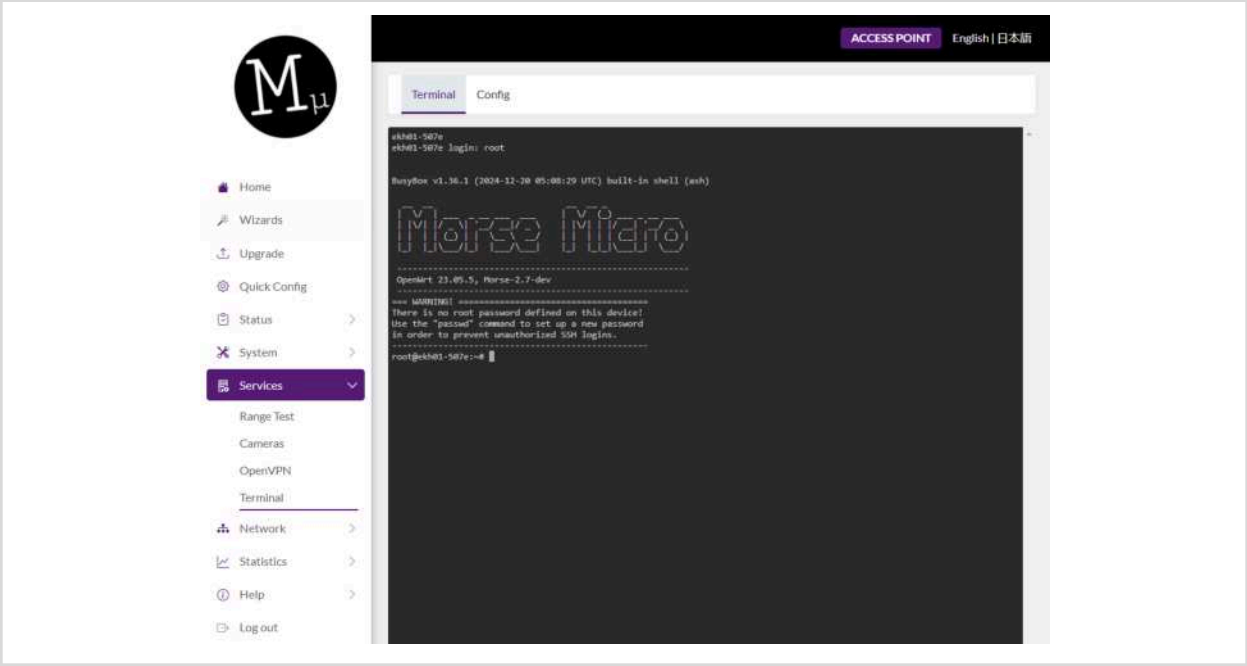


Figure: The Terminal page

Note: The in-browser terminal does not work over HTTPS. To use it, follow the browser's warning link to disable this security feature.

From the command prompt, you will have access to the standard Linux utilities included in OpenWrt and those already mentioned via the Diagnostics page, as well as some other useful programs such as:

Utility	Purpose
`morse_cli`	Low level access to information from and settings on the Morse Micro HaLow chip.
`wavemon`	Terminal graphical program to monitor Wi-Fi signal strength and performance.
`nano`	Text editor, including syntax highlighting of UCI files. You may also use vi.
`tmux`	Terminal multiplexer, allowing persistent sessions and windows.

Table: Useful Command Line Interface tools

7.6.2 OpenVPN

This tool is included from vanilla OpenWrt and provides an easy-to-use interface for configuring and managing OpenVPN client connections (see their documentation).

7.7 Network

7.7.1 Wireless Page

This page provides access to more advanced radio and network controls. As is pictured below in the Wireless Overview section there are 3 radios available where the **Morse Micro USB-MM810x 802.11ah** row represents the Morse Micro EKH19's HaLow USB dongle shown in this diagram.

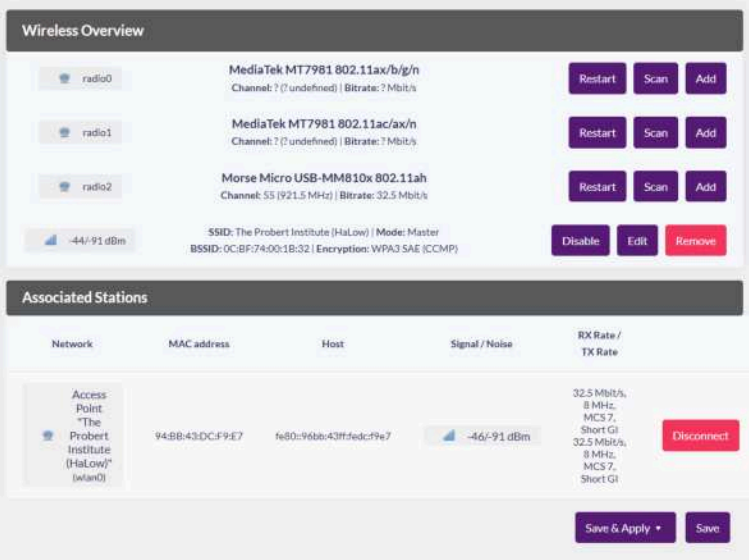


Figure: Wireless page

In the final row of the Wireless Overview section is a HaLow wireless network configured as an Access Point called **The Probert Institute (HaLow)** which is operating on the HaLow radio (**radio2**) above it. By clicking the **Edit** button on this wireless network a dialog box will appear with advanced configuration options, explored further in Additional Configuration Parameters.

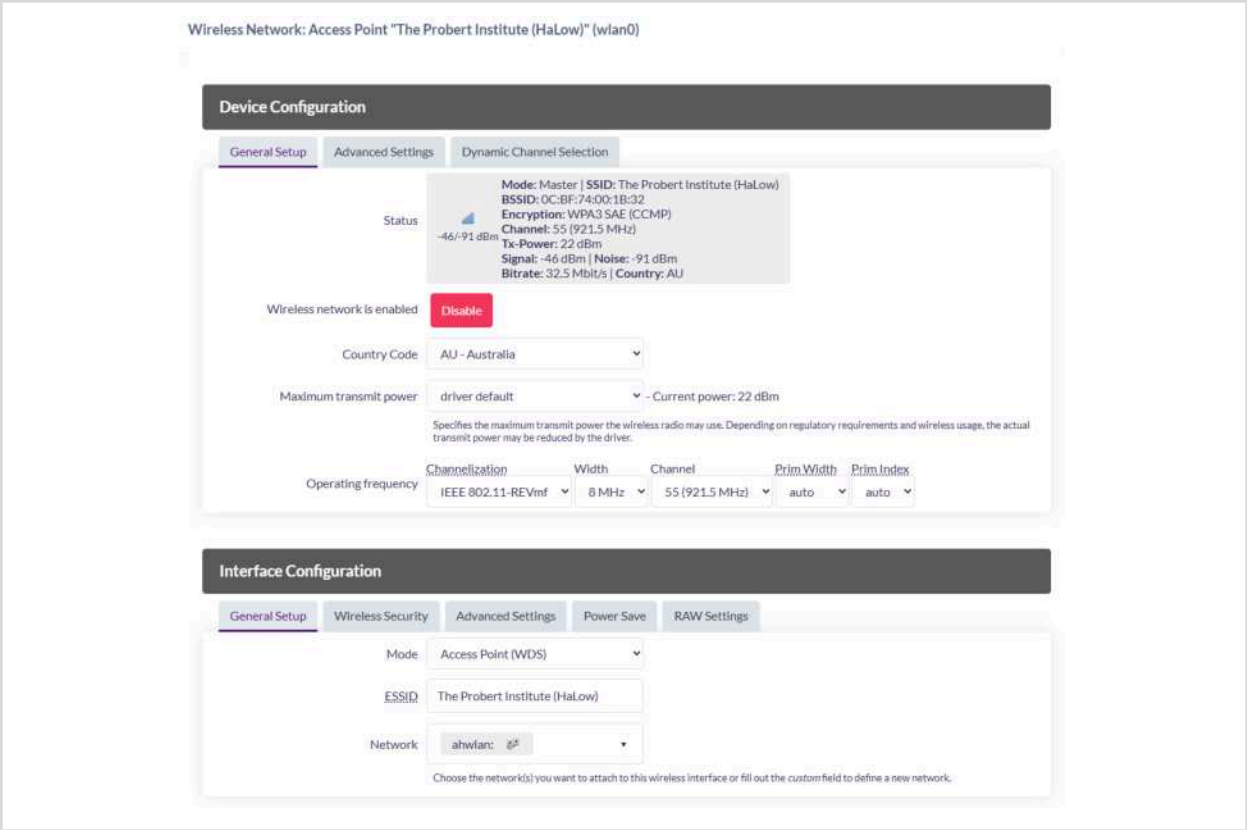


Figure: Wireless page wireless network advanced configuration

7.8 Statistics

7.8.1 Morse Page

This page provides processor core statistics on the Morse chip. **Read** displays the current statistics of a given core, whilst **Reset** sets counters back to zero. The underlying information on this page is gathered via the `morse_cli` command, which is also available via the CLI.

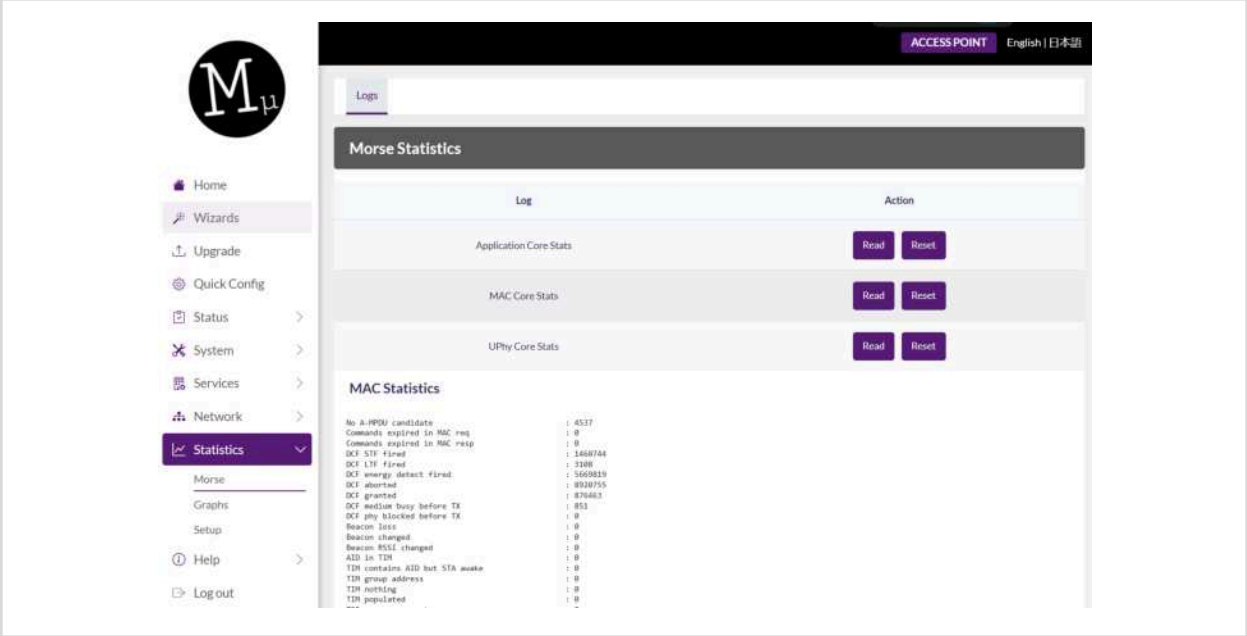


Figure: Morse statistics page

7.9 Help

7.9.1 Regulatory Information

This page provides regulatory details for the selected country, including permissible 802.11ah HaLow parameters like channel frequency, transmission power, and duty cycle limits. It also indicates which channels are supported on your device, helping ensure compliance with local wireless regulations when planning deployments.

8 Additional Configuration Parameters

Some advanced configurations are useful to control HaLow behavior (particularly during certifications) and are documented here. Some may only be available via the CLI. If unsure, it is best not to modify default values unless advised by a Morse Micro engineer to do so.

8.1 Disable AMPDU

8.1.1 Via UI

AMPDU can be disabled (or re-enabled) by navigating to the Wireless page in the Network sub-menu and then choosing **Edit** beside the HaLow network. De-select the AMPDU input in the **Advanced Settings** tab of the **Device Configuration** section:

8.1.2 Via CLI

AMPDU can be disabled using the following command:

```
morse_cli -i wlan0 ampdu disable
```

Note: Configurations made via the CLI are not persistent across reboots. To ensure your settings are saved permanently, please configure them through the UI.

8.2 Fragmentation Threshold

8.2.1 Via UI

In the same configuration section as described above for AMPDU, there is an option for configuring the fragmentation threshold. To disable this feature enter 'off' into the field, otherwise enter the number of bytes beyond which fragmentation should occur.

8.2.2 Via CLI

The fragmentation threshold can be set with the `iw` tool:

```
iw phy <phyname> set frag <fragmentation threshold|off>
```

Where the `<phyname>` is provided by:

```
iw list | grep wiphy
```

Which might return something like `wiphy phy1` where `phy1` is the `<phyname>`. The integer following phy increments every time the driver is (re)loaded.

Note: Configurations made via the CLI are not persistent across reboots. To ensure your settings are saved permanently, please configure them through the UI.

8.3 Station Inactivity Limit

Sets the maximum number of seconds a wireless client (station) can remain inactive before the AP considers it disconnected or inactive.

8.3.1 Via UI

Navigate to the Wireless page in the Network sub-menu and then choose **Edit** beside the HaLow network. Set the number of seconds to wait before removing a station in the **Station inactivity limit** input in the **Advanced Settings** tab of the **Interface Configuration** section:

8.4 Unified Scaling Factor / Unscaled Interval

8.4.1 Via UI

The Forced Listen Interval can be configured by navigating to the Wireless page in the Network sub-menu and then choosing **Edit** beside the HaLow network. Set the **Forced Listen Interval** input in the **Advanced Settings** tab of the **Device Configuration** section:

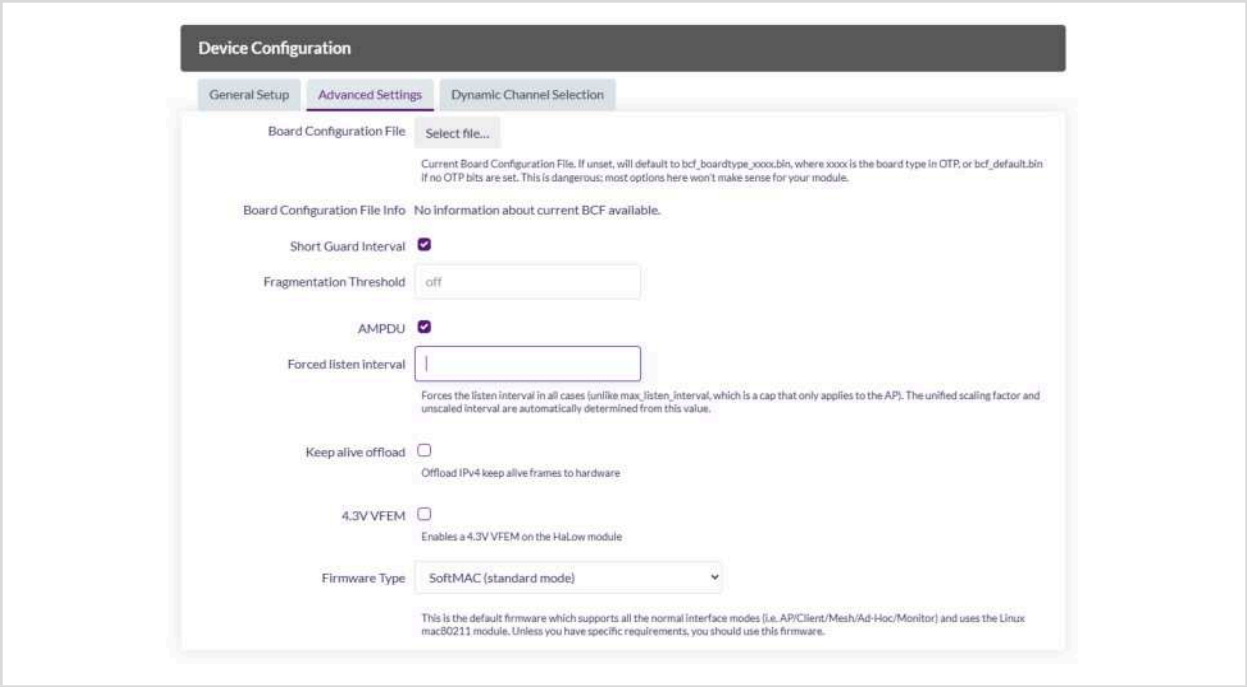


Figure: Force Listen Interval configuration

8.4.2 Via CLI

The Unified Scaling Factor and Unscaled Interval and must be set together, with the `morse_cli` tool using the command:

```
morse_cli -i wlan0 li <unscaled interval> <unified scaling factor>
```

Where `<unscaled interval>` multiplied by `<unified scaling factor>` must be less than or equal to the integer value 65536.

Note: Configurations made via the CLI are not persistent across reboots. To ensure your settings are saved permanently, please configure them through the UI.

8.5 DTIM Interval

Specifies the DTIM (Delivery Traffic Indication Message) period to use, defined as the number of consecutive beacon intervals between DTIM transmissions.

8.5.1 Via UI

DTIM Interval can be configured by navigating to the Wireless page in the Network sub-menu and then choosing **Edit** beside the HaLow network. Set the **DTIM Interval** input in the **Advanced Settings** tab of the **Interface Configuration** section:

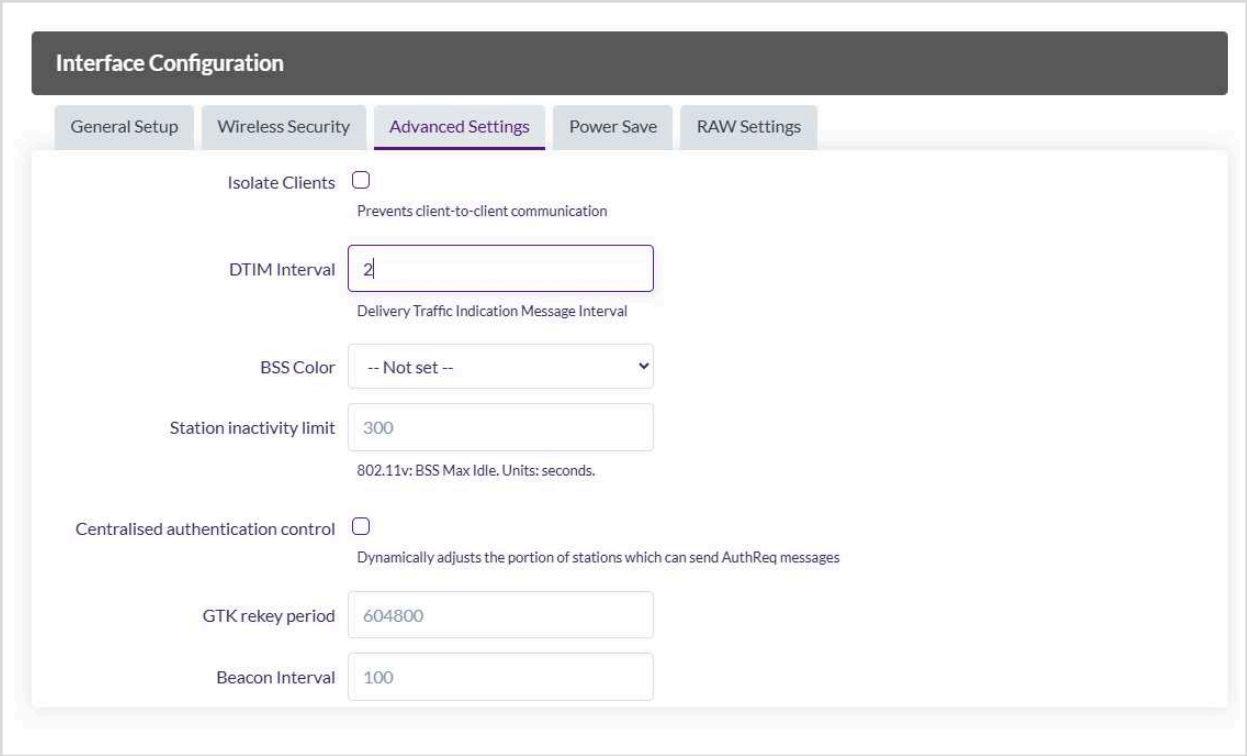


Figure: DTIM Interval configuration

8.6 Beacon Interval

Defines how often beacons are broadcast, measured in time units where 1 TU is equal to 1.024 milliseconds.

8.6.1 Via UI

Beacon interval can be configured by navigating to the **Network** >**Wireless** page and then choose Edit beside the HaLow network. Use the **Beacon Interval** in the **Advanced Settings** tab of the **Interface Configuration** section:

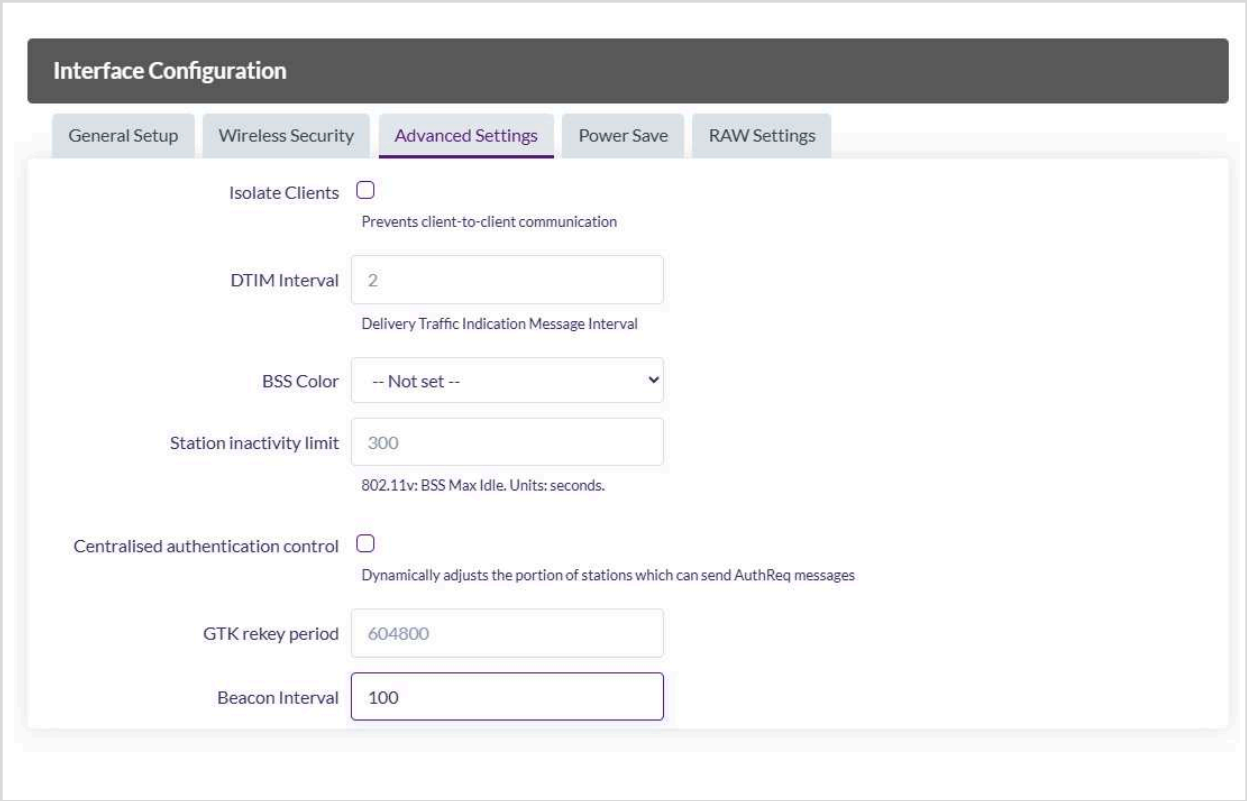


Figure: Beacon Interval configuration

8.7 BSS Color

On an Access Point, assigning your network a unique BSS color (a numeric identifier in the headers) allows stations to quickly distinguish between sources of overlapping HaLow traffic reducing contention and interference.

8.7.1 Via UI

Navigate to the **Network** > **Wireless** page and then choose **Edit** beside the HaLow network. Use the **BSS Color** list in the **Advanced Settings** tab of the **Interface Configuration** section:

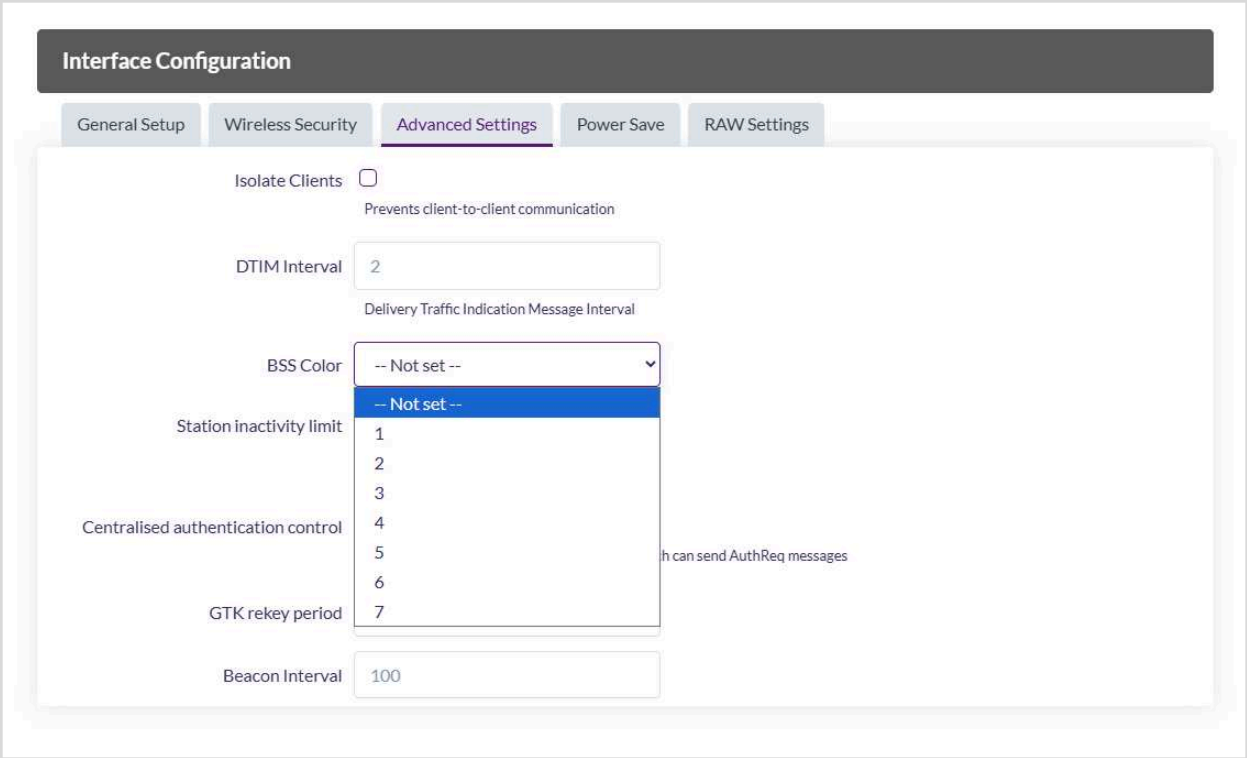


Figure: BSS Color setting

8.7.2 Via CLI

BSS color can be configured using the following command:

```
morse_cli -i wlan0 bsscolor <value>
```

Where `<value>` is an integer from 0 to 7.

Note: Configurations made via the CLI are not persistent across reboots. To ensure your settings are saved permanently, please configure them through the UI.

8.8 Thin LMAC

Thin LMAC mode allows the firmware running in AP mode to support a large number of associations. This is achieved by reducing the per-station table memory from 88 bytes to 8 bytes inside the chip and moving them to the upper layers of the stack onto the host processor.

8.8.1 Via UI

To enable Thin LMAC through UI navigate to **Network** >**Wireless** and choose **Edit** beside the HaLow network. Select the **Thin LMAC** Firmware Type option in Advanced Settings tab of the **Device Configuration** section:

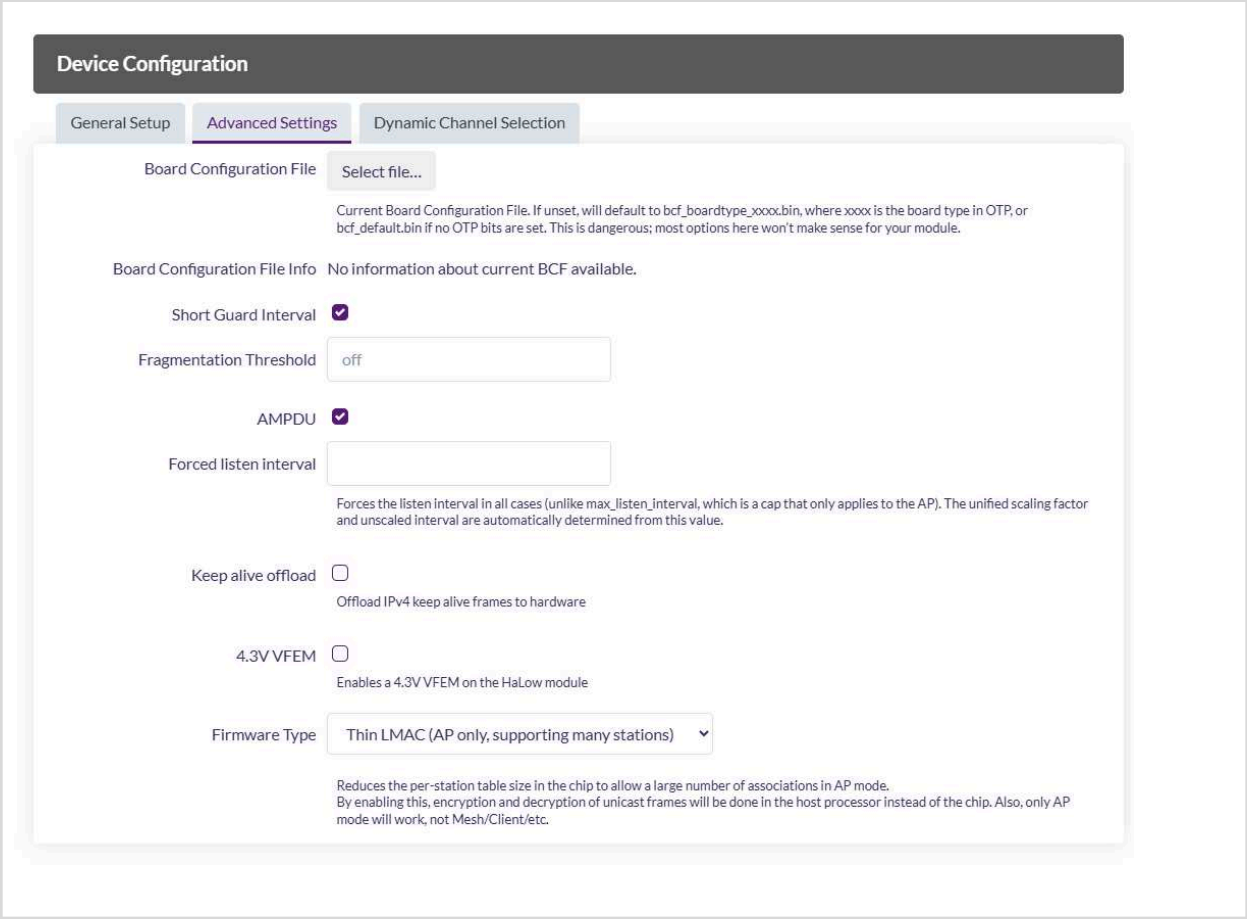


Figure: Thin LMAC firmware enabled

8.8.2 Via CLI

Enable thin LMAC from the command line with the following commands:

```
uci set wireless.radio1.thin_lmac='1'
uci commit
reload_config
```

Disable thin LMAC by setting the above option back to 0 or deleting it from UCI.

8.8.3 Thin LMAC Considerations

Since the goal of thin LMAC is to allow a large number of stations, there are some further suggested changes that may be helpful to reduce unnecessary traffic. These include:

- Disabling IPv6
- Reducing ARP garbage collection frequency
- Increasing ARP table entry timeout
- Disabling unnecessary ARP responses

8.9 Other HaLow settings

Other advanced settings are available within the text files found at `/etc/config/`. Generic options are defined in the OpenWrt documentation: https://openwrt.org/docs/guide-user/network/wifi/basic.

8.10 morse_cli

`morsectrl` is a command line utility that allows low-level access and control of the Morse radio chip. It is not intended to be included on consumer devices.

`morse_cli` contains a subset of `morsectrl` commands and is used by `netifd` to configure the radio. It is intended to be shipped on consumer devices.

On all the devices mentioned in Device Setup, both of these utilities are included. For more information about their available options, refer to their respective command help messages via `-h`.

Note: Configurations made via the CLI are not persistent across reboots. To ensure your settings are saved permanently, please configure them through the UI.

8.11 Dynamic Channel Selection (DCS)

Dynamic Channel Selection (DCS) enables an Access Point (AP) to periodically scan other channels while maintaining an active BSS and automatically switch to a less congested or higher-quality channel. This improves long-term network performance, especially in RF-congested environments.

8.11.1 Algorithms

DCS uses channel quality metrics collected during periodic scans to evaluate whether candidate channels may provide better performance. These evaluations are performed by algorithms - **EWMA** or **Sample and Hold** - which determine whether a channel switch should be triggered based on configurable thresholds and timing behaviour. The 2 algorithms available in Morse OpenWrt include:

- **EWMA** uses an exponential moving average to continuously track channel quality.
- **Sample and Hold** accumulates data over a period before making decisions.

8.11.2 Via UI

To access DCS settings:

- Go to **Network** > **Wireless**
- Click **Edit** on your HaLow AP interface
- Open the **Dynamic Channel Selection** tab in the Device Configuration section.

The following parameters are available for your configuration:

- **Algorithm** used to evaluate candidate channel quality.
- **EWMA Alpha** (EWMA only) controls how much weight is placed on new vs historical channel scores. Higher values bias recent measurements more heavily.
- **Rounds** sets the number of scan rounds required to confirm a better channel. For EWMA, this is the number of consecutive better rounds; for Sample and Hold, it's the number of rounds over which scores are averaged.
- **Threshold** specifies how much better a candidate channel must be (as a percentage) to trigger a switch.
- **Scan interval (s)** how long to wait between individual channel scans in a round.
- **Round interval (s)** how long to wait between entire scan rounds.

9 UI Configuration Architecture

This section outlines how changes to configuration in the UI are applied to the system.

In the UI many settings are often tied to **UCI**, the **Unified Configuration Interface**, which is OpenWrt's centralized system for managing device configuration. The UI generally reads and writes from standard configuration sections in `/etc/config/` via the `LuCI.uci` API. On configuration pages such as Quick Config, when users click **Save & Apply**, the changes are written to the device's UCI.

If you only click the **Save** button instead of **Save & Apply**, the changes will be staged and can be applied later by clicking on the **UNSAVED CHANGES** indicator on the banner at the top of the screen. The dialog that pops up will also show you the exact UCI changes which will be propagated.



Figure: The UNSAVED CHANGES indicator

After this the network service daemon, `netifd`, examines changed UCI configurations upon reload and calls the necessary handler scripts to update the affected components. For example, a UCI configuration like `wireless.wifi-device`, `netifd` invokes wireless protocol handlers located in `/lib/netifd/wireless/*.sh`. For changes to Morse Micro HaLow devices where the UCI configuration includes `type=morse`, `netifd` will load `/lib/netifd/wireless/morse.sh`.

This protocol handler carries out the following steps:

- Parses sections with a Morse type of `wifi-device` in `/etc/config/wireless`
- Kills `hostapd_s1g` and `wpa_supplicant_s1g`
- Tears down the HaLow configured interfaces
- Rebuilds any morse module parameters - e.g. region information
- Reloads the morse driver module if parameters have changed
- Brings up the HaLow interface
- Creates appropriate `hostapd` or `wpa_supplicant` configuration files.
- Starts `hostapd_s1g` or `wpa_supplicant_s1g` as required.

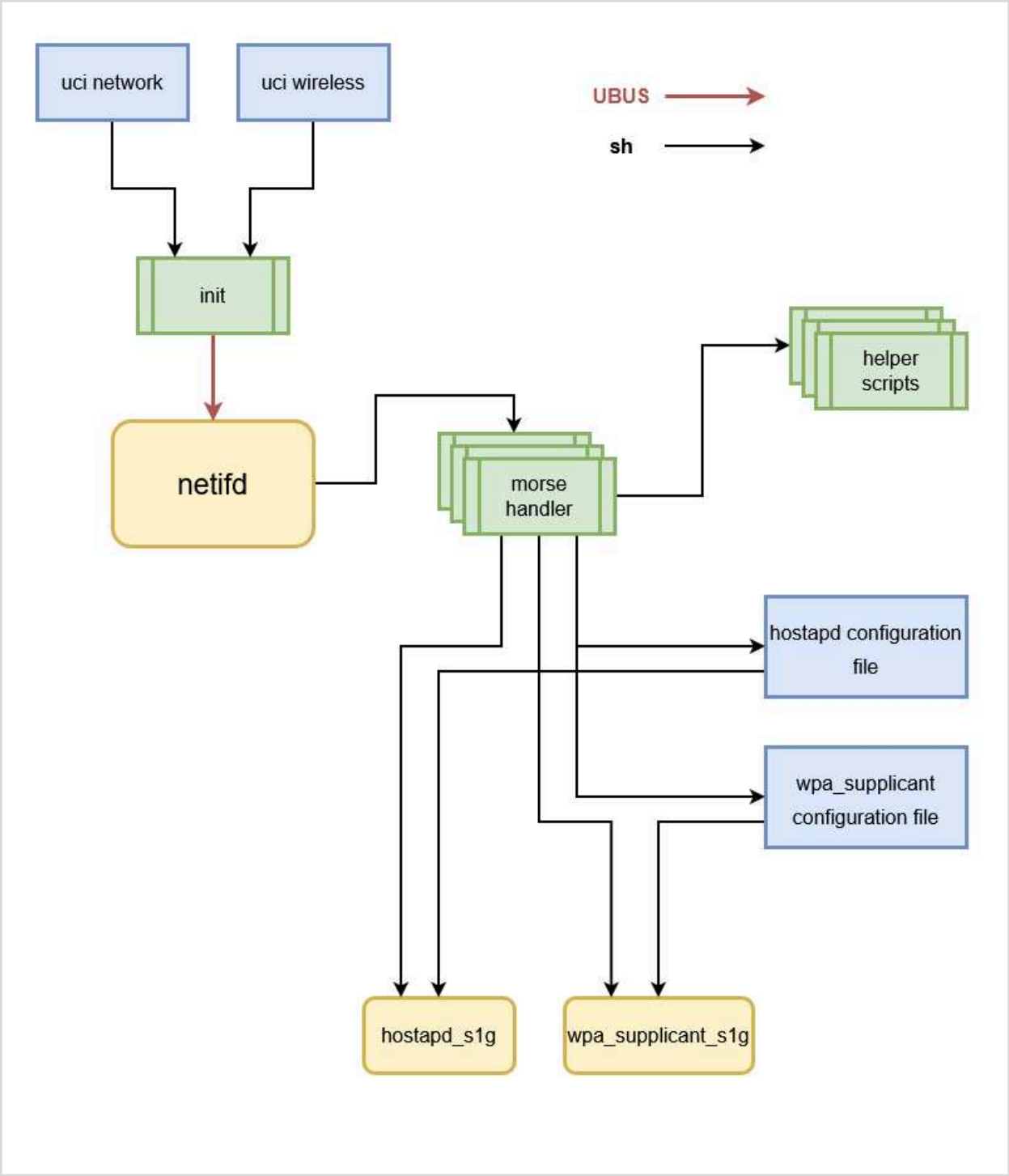


Figure: UI configuration architecture flowchart

10 Troubleshooting

10.1 Restoring Factory Settings

This section outlines how to get the device back to a default configuration in different situations. All firmware releases use SquashFS with an overlay which allows for full factory resets.

10.1.1 Access to Web UI is Available

In this scenario, you can login to the device and go to the Backup / Flash Firmware page under the System sub menu. Selecting **Reset to defaults** will reset the device's configuration and reboot.

10.1.2 SSH Access is Available

In this scenario, you can login to the device via ssh (`ssh root@<ipaddress>`) and run the following command at the prompt:

```
firstboot -y && reboot
```

10.1.3 No Network Access - EKH01

This scenario can occur when the IP address of the device has been changed, and it is not obvious what the address is. The quickest method here is to remove the SD card and write a new firmware image to it using a tool such as Balena Etcher or the Raspberry Pi Imager.

10.1.4 No network access (Option 1) - EKH03

This scenario can occur when the IP address of the device has been changed, and the user is unable to identify it via the lease list on their DHCP server. Using a suitable object, press and hold the reset button, shown in the EKH03 diagram, for more than 10 seconds to reset the device to factory defaults. It will first start flashing slowly green (for a normal reset) after 5 seconds, then after 10 seconds will quickly flash yellow. Release the button once it's flashing yellow. For a graphical view of this behavior see the EKH03 LED Behavior section.

10.1.5 No network access (Option 2) - EKH03

This scenario requires using a serial console cable to access the device. The EKH03 diagram shows the location of the USB port on the EKH03 where a 3.3V TTL serial USB cable can be used to connect a computer to this port. Once connected, a suitable terminal emulator program will be

needed to connect to it, such as PuTTY on Windows or picocom on Linux. Once connected to the serial console, run the following command to reset the configuration.

10.2 Firmware Update Failures

Occasionally a platform name is updated, which can result in an error during upgrade e.g. “The uploaded image does not contain a supported format”. Before proceeding, check that the **Supported devices** line matches the device revision; the device revision is printed on the case, and on a sticker on the case.

Pictured below, the OpenWrt image currently running on the device is for an EKH01 (purple highlighting) whilst the upgrade has a name device change and is intended for a new device name: EKH01-03 (blue highlighting).

Once you have verified that the upgrade image matches your device the **Force Upgrade** button can be used and warnings will not show again for future upgrades.

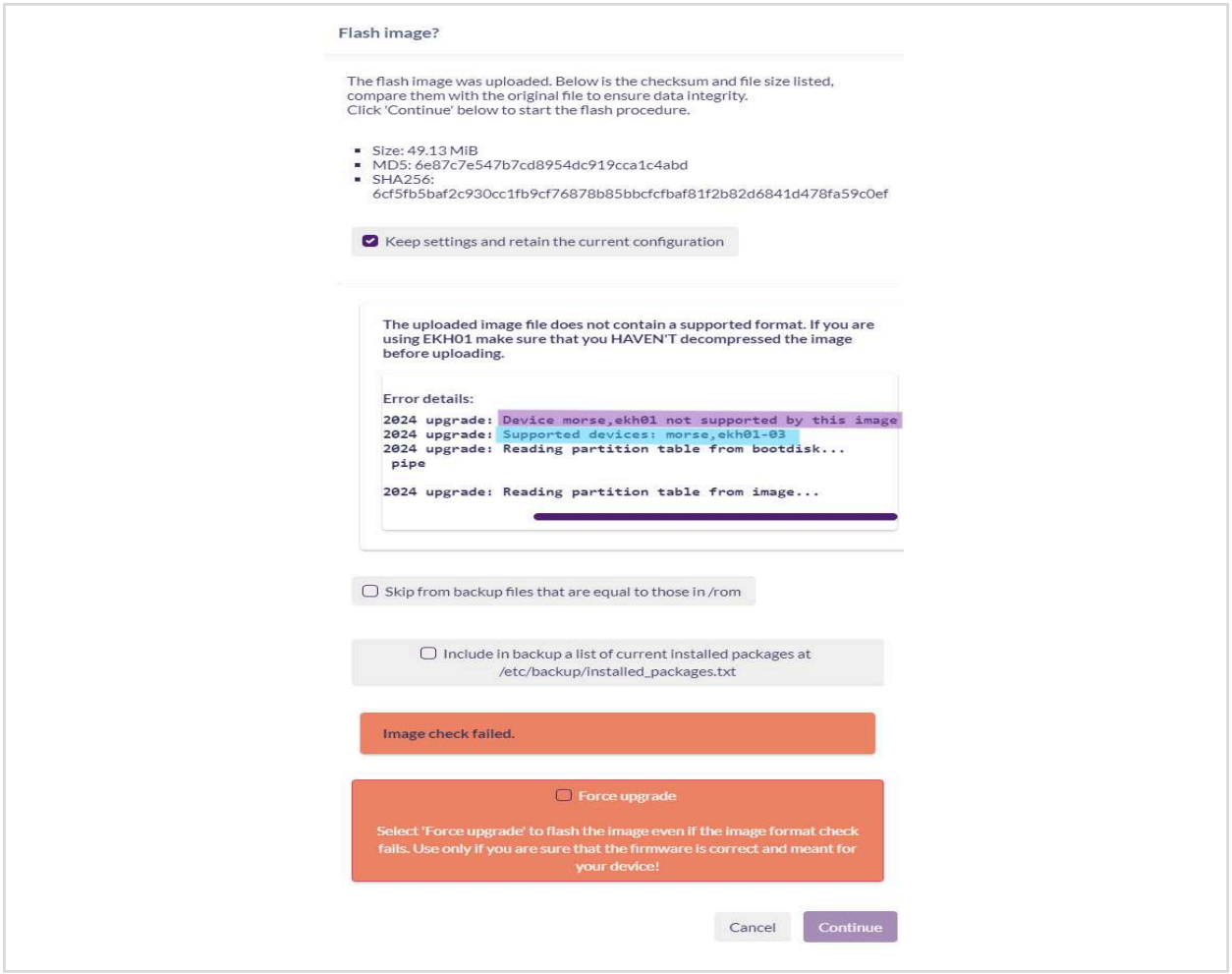


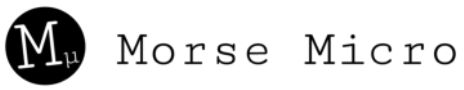
Figure: Device not supported by this image upgrade error

11 Revision History

Release Number	Release Date	Release Notes
01	12/01/2021	Initial release
02	12/02/2022	Update for firmware release 1.3
03	04/03/2022	IPERF Traffic Setup Added Tools > HaLow Firmware Upgrade
04	05/10/2022	Updated for the LuCI interface Added in EKH01
05	10/10/2022	Improved formatting and reworded some sections for clarity. Added UI Configuration architecture
06	20/10/2022	Added example of how to run wavemon for basic HaLow testing
07	18/10/2022	Add description of key setup scenarios, and refactored configuration to match these. Removed references to custom configurations, and manual configuration except where not available in UI. Other general improvements
08	22/11/2022	Updated device images
09	12/12/2022	Updated formatting and cover page image
10	6/01/2023	Updated for UCI configuration Updated default IP address to 10.42.0.1
11	27/02/2023	Correct some typos
12	02/06/2023	Update for new HaLow configuration page Update for new EasyMesh feature Update for new Video UI feature Update for adding internet connectivity
13	03/11/2023	General update and 1st release to Doc. Control
14	12/12/2023	Updated for 2.4.4 release
15	7/03/2024	Updated for 2.5.0 release
16	21/03/2024	Updated for 2.5.2 release

17	28/03/2024	Updated LED flash pattern for button presses (section 2.2.1)
18	02/08/2024	Updated for 2.6 release
19	17/12/2024	Updated for 2.7 release Updated EKH03 LED changes Added LTE features section
20	20/12/2024	Updated for 2.7.1 release Added Range Testing section
21	14/01/2025	Updated for 2.7.2 release Add monitor mode guide
22	14/04/2025	Updated for 2.8.3 release Added in EKH19 Add Default Web UI IP Addresses section
23	20/05/2025	Updated for 2.8.5 release
24	09/07/2025	Updated for 2.9 release
25	23/07/2025	Add MBSSID section Added DCS section
26	31/10/2025	Formatting update
27	5/11/2025	Updated for 2.11 release
28	16/02/2026	Updated for 2.12 release
29	28/05/2026	Separated guide versioning from OpenWrt - Updated for 3.0 and 3.1 releases Added Native S1G section
30	22/07/2026	Added HaLow Diagnostics section

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